

Question 1Incorrect

Which of the following is a perspective of the AWS Cloud Adoption Framework (AWS CAF)?

Correct answer

Business

Your answer is incorrect

Architecture

Product

Process

Overall explanation

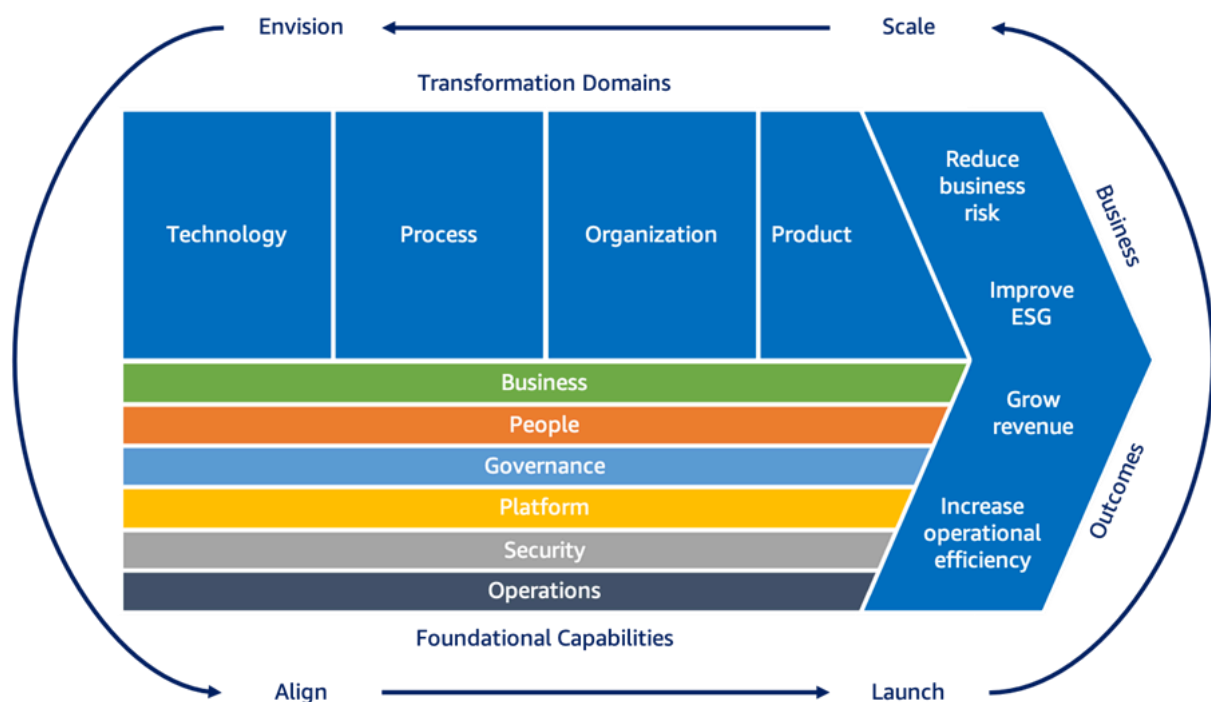
Correct option:

Business

The AWS Cloud Adoption Framework (AWS CAF) leverages AWS experience and best practices to help you digitally transform and accelerate your business outcomes through innovative use of AWS. AWS CAF identifies specific organizational capabilities that underpin successful cloud transformations.

AWS CAF groups its capabilities in six perspectives: Business, People, Governance, Platform, Security, and Operations.

AWS Cloud Adoption Framework (CAF) Overview:



via -

<https://aws.amazon.com/blogs/aws/aws-cloud-adoption-framework-caf-3-0-is-now-available/>

Incorrect options:

Process

Product

Architecture

These three options contradict the explanation provided above, so these options are incorrect.

References:

<https://aws.amazon.com/cloud-adoption-framework/>

<https://aws.amazon.com/blogs/aws/aws-cloud-adoption-framework-caf-3-0-is-now-available/>

Domain

Cloud Concepts

Question 2Correct

The DevOps team at an IT company wants to centrally manage its servers on AWS Cloud as well as on-premise data center so that it can collect software inventory, run commands, configure and patch servers at scale. As a Cloud Practitioner, which AWS service would you recommend for this use-case?

AWS CloudFormation

AWS Config

AWS OpsWorks

Your answer is correct

AWS Systems Manager

Overall explanation

Correct option:

AWS Systems Manager

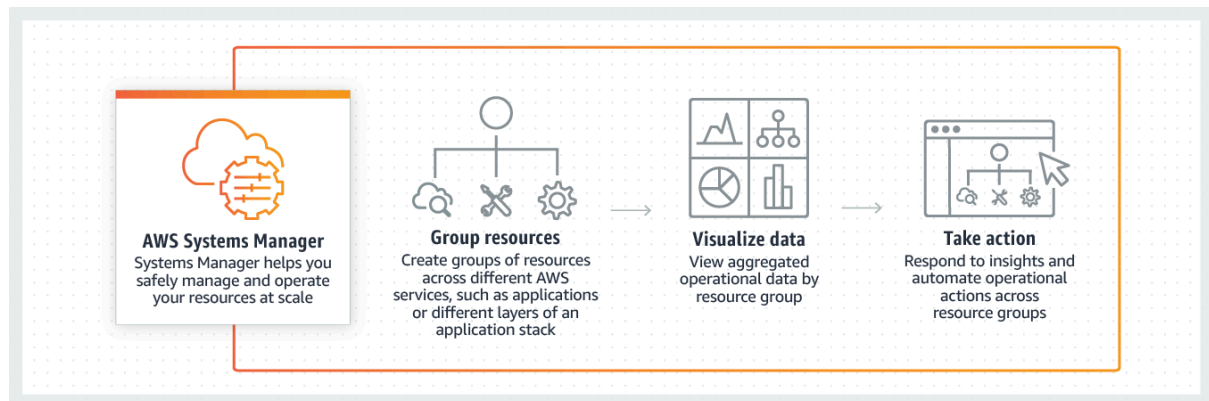
AWS Systems Manager gives you visibility and control of your infrastructure on AWS. Systems Manager provides a unified user interface so you can view operational data from multiple AWS services and allows you to automate operational tasks such as

collecting software inventory, running commands, managing patches, and configuring servers across AWS Cloud as well as on-premises infrastructure.

AWS Systems Manager offers utilities for running commands, patch-management and configuration compliance:



via - <https://aws.amazon.com/systems-manager/faq/>



via - <https://aws.amazon.com/systems-manager/>

Incorrect options:

AWS OpsWorks - AWS OpsWorks is a configuration management service that provides managed instances of Chef and Puppet. AWS OpsWorks lets you use Chef and Puppet to automate how servers are configured, deployed and managed across your Amazon EC2 instances or on-premises compute environments. You cannot use AWS OpsWorks for collecting software inventory and viewing operational data from multiple AWS services.

AWS CloudFormation - AWS CloudFormation allows you to use programming languages or a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all Regions and accounts. Think infrastructure as code; think AWS CloudFormation. You cannot use AWS CloudFormation for running commands or managing patches on servers.

AWS Config - AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. AWS Config continuously monitors and records your AWS resource configurations and allows you to automate the

evaluation of recorded configurations against desired configurations. You cannot use AWS Config for running commands or managing patches on servers.

References:

<https://aws.amazon.com/systems-manager/>

<https://aws.amazon.com/systems-manager/faq/>

Domain

Technology

Question 3Incorrect

Bob and Susan each have an AWS account in AWS Organizations. Susan has five Reserved Instances (RIs) of the same type and Bob has none. During one particular hour, Susan uses three instances and Bob uses six for a total of nine instances on the organization's consolidated bill.

Which of the following statements are correct about consolidated billing in AWS Organizations? (Select two)

Correct selection

AWS bills five instances as Reserved Instances, and the remaining four instances as regular instances

Your selection is incorrect

AWS bills three instances as Reserved Instances (RI), and the remaining six instances as regular instances

Bob receives the cost-benefit from Susan's Reserved Instance (RI) only if he launches his instances in the same AWS Region where Susan purchased her Reserved Instances (RI)

Bob does not receive any cost-benefit since he hasn't purchased any Reserved Instance (RI). If his account has even one RI, then the cost-benefit from Susan's account is also added to his account

Your selection is correct

Bob receives the cost-benefit from Susan's Reserved Instances (RI) only if he launches his instances in the same Availability Zone (AZ) where Susan purchased her Reserved Instances

Overall explanation

Correct options:

Bob receives the cost-benefit from Susan's Reserved Instances (RI) only if he launches his instances in the same Availability Zone (AZ) where Susan purchased her Reserved Instances

Bob receives the cost-benefit from Susan's Reserved Instances (RI) only if he launches his instances in the same Availability Zone (AZ) where Susan purchased her Reserved Instances. For example, if Susan specifies us-west-2a when she purchases her Reserved Instances, Bob must specify us-west-2a when he launches his instances to get the cost-benefit on the organization's consolidated bill. However, the actual locations of Availability Zones (AZs) are independent of one account to another. For example, the us-west-2a Availability Zone (AZ) for Bob's account might be in a different location than the location for Susan's account.

AWS bills five instances as Reserved Instances, and the remaining four instances as regular instances

Since Susan has five Reserved Instances (RIs), AWS bills five instances as Reserved Instances, and the remaining four instances as regular instances.

Incorrect options:

AWS bills three instances as Reserved Instances (RI), and the remaining six instances as regular instances - This option contradicts the explanation provided above, so it's incorrect.

Bob does not receive any cost-benefit since he hasn't purchased any Reserved Instance (RI). If his account has even one RI, then the cost-benefit from Susan's account is also added to his account - For billing purposes, the consolidated billing feature of AWS Organizations treats all the accounts in the organization as one account. This means that all accounts in the organization can receive the hourly cost-benefit of Reserved Instance (RI) that are purchased by any other account.

Bob receives the cost-benefit from Susan's Reserved Instance (RI) only if he launches his instances in the same AWS Region where Susan purchased her Reserved Instances (RI) - As discussed above, this statement is incorrect. Bob receives the cost-benefit from Susan's Reserved Instances only if he launches his instances in the same Availability Zone (AZ) where Susan purchased her Reserved Instance (RI).

Reference:

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/consolidatedbilling-other.html>

Domain

Billing and Pricing

Question 4Incorrect

Which of the following is the best practice for application architecture on AWS Cloud?

Build tightly coupled components

Your answer is incorrect

Use synchronous communication between components

Correct answer

Build loosely coupled components

Build monolithic applications

Overall explanation

Correct option:

Build loosely coupled components

AWS Cloud recommends microservices as an architectural and organizational approach to software development where software is composed of small independent services that communicate over well-defined APIs. These services are owned by small, self-contained teams.

Microservices architectures make applications easier to scale and faster to develop, enabling innovation and accelerating time-to-market for new features. Each service can be considered as a loosely coupled component of a bigger system. You can use services like SNS or SQS to decouple and scale microservices.

Microservices Overview:

What are Microservices?

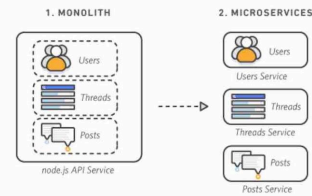
Microservices are an architectural and organizational approach to software development where software is composed of small independent services that communicate over well-defined APIs. These services are owned by small, self-contained teams.

Microservices architectures make applications easier to scale and faster to develop, enabling innovation and accelerating time-to-market for new features.

Monolithic vs. Microservices Architecture

With monolithic architectures, all processes are tightly coupled and run as a single service. This means that if one process of the application experiences a spike in demand, the entire architecture must be scaled. Adding or improving a monolithic application's features becomes more complex as the code base grows. This complexity limits experimentation and makes it difficult to implement new ideas. Monolithic architectures add risk for application availability because many dependent and tightly coupled processes increase the impact of a single process failure.

With a microservices architecture, an application is built as independent components that run each application process as a service. These services communicate via a well-defined interface using lightweight APIs. Services are built for business capabilities and each service performs a single function. Because they are independently run, each service can be updated, deployed, and scaled to meet demand for specific functions of an application.



Breaking a monolithic application into microservices

via -

<https://aws.amazon.com/blogs/compute/understanding-asynchronous-messaging-for-microservices/>

Incorrect options:

Build tightly coupled components

Build monolithic applications

With monolithic architectures, all processes are tightly coupled and run as a single service. This means that if one process of the application experiences a spike in demand, the entire architecture must be scaled. Monolithic architectures add risk for application availability because many dependent and tightly coupled processes increase the impact of a single process failure. So both these options are incorrect.

Use synchronous communication between components - Synchronous between applications can be problematic if there are sudden spikes of traffic. You should use SNS or SQS to decouple your application components.

Reference:

<https://aws.amazon.com/blogs/compute/understanding-asynchronous-messaging-for-microservices/>

Domain

Cloud Concepts

Question 5Incorrect

A social media analytics company wants to migrate to a serverless stack on AWS.

Which of the following scenarios can be handled by AWS Lambda? (Select two)

Your selection is incorrect

You can install low latency databases on AWS Lambda

You can install Container Services on AWS Lambda

AWS Lambda can be used to store sensitive environment variables

Correct selection

AWS Lambda can be used to execute code in response to events such as updates to DynamoDB tables

Your selection is correct

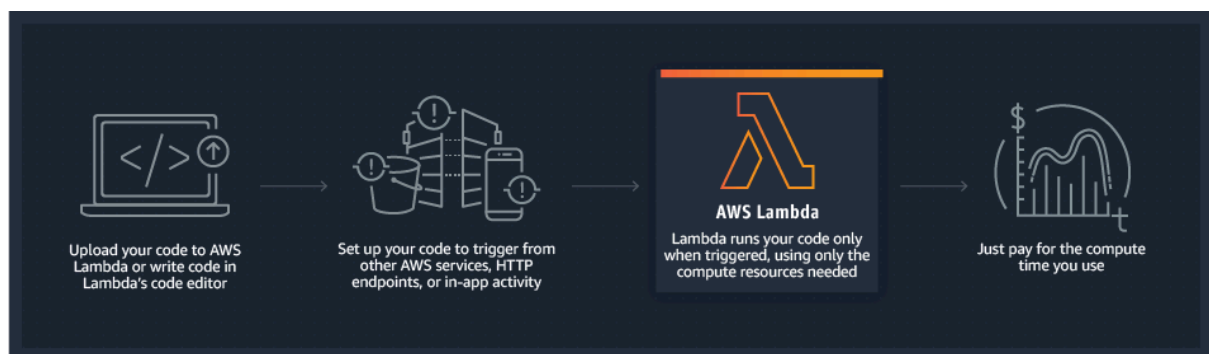
AWS Lambda can be used for preprocessing of data before it is stored in Amazon S3 buckets

Overall explanation

Correct options:

AWS Lambda lets you run code without provisioning or managing servers (Lambda is serverless). With Lambda, you can run code for virtually any type of application or backend service - all with zero administration. Just upload your code and Lambda takes care of everything required to run and scale your code with high availability. This functionality makes it an extremely useful service capable of being a serverless backend for websites, data preprocessing, real-time data transformations when used with streaming data, etc.

How AWS Lambda Works:



via - <https://aws.amazon.com/lambda/>

AWS Lambda can be used to execute code in response to events such as updates to DynamoDB tables

AWS Lambda can be configured to execute code in response to events, such as changes to Amazon S3 buckets, updates to an Amazon DynamoDB table, or custom events generated by your applications or devices.

AWS Lambda can be used for preprocessing of data before it is stored in Amazon S3 buckets

AWS Lambda can be used to run preprocessing scripts to filter, sort or transform data before sending it to downstream applications/services.

Incorrect options:

You can install low latency databases on AWS Lambda - AWS Lambda is serverless, so the underlying hardware and its working is not exposed to the customer. Installing software is not possible since we do not have access to the actual physical server on which Lambda executes the code.

You can install Container Services on AWS Lambda - As discussed above, AWS Lambda cannot be used for installing any software, since the underlying hardware/software might change for each request. But, it is possible to set an environment with necessary libraries when running scripts on Lambda.

AWS Lambda can be used to store sensitive environment variables - AWS Lambda is not a storage service and does not offer capabilities to store data. However, it is possible to read and decrypt/encrypt data using scripts in Lambda.

Reference:

<https://aws.amazon.com/lambda/>

Domain

Technology

Question 6Incorrect

Which of the following is the MOST cost-effective Amazon Elastic Compute Cloud (Amazon EC2) instance purchasing option for short-term, spiky and critical workloads on AWS Cloud?

Reserved Instance (RI)

Your answer is incorrect

Dedicated Host

Spot Instance

Correct answer

On-Demand Instance

Overall explanation

Correct option:

On-Demand Instance

An On-Demand Instance is an instance that you use on-demand. You have full control over its lifecycle — you decide when to launch, stop, hibernate, start, reboot, or terminate it. There is no long-term commitment required when you purchase On-Demand Instances. There is no upfront payment and you pay only for the seconds that your On-Demand Instances are running. There is no need for a long-term purchasing commitment. The price per second for running an On-Demand Instance is fixed. On-demand instances cannot be interrupted. Therefore On-Demand instances are the best fit for short-term, spiky and critical workloads.

Amazon EC2 Pricing Options Overview:

On-Demand

With On-Demand instances, you pay for compute capacity by the hour or the second depending on which instances you run. No longer-term commitments or upfront payments are needed. You can increase or decrease your compute capacity depending on the demands of your application and only pay the specified per hourly rates for the instance you use.

On-Demand instances are recommended for:

- Users that prefer the low cost and flexibility of Amazon EC2 without any up-front payment or long-term commitment
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

[See On-Demand pricing »](#)

Spot instances

Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. [Learn More](#).

Spot instances are recommended for:

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with urgent computing needs for large amounts of additional capacity

[See Spot pricing »](#)

Savings Plans

Savings Plans are a flexible pricing model that offer low prices on EC2 and Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term.

Dedicated Hosts

A Dedicated Host is a physical EC2 server dedicated for your use. Dedicated Hosts can help you reduce costs by allowing you to use your existing server-bound software licenses, including Windows Server, SQL Server, and SUSE Linux Enterprise Server (subject to your license terms), and can also help you meet compliance requirements. [Learn more](#).

- Can be purchased On-Demand (hourly).
- Can be purchased as a Reservation for up to 70% off the On-Demand price.

[See Dedicated pricing »](#)

Reserved Instances

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them.

For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. See [How to Purchase Reserved Instances](#) for more information.

Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

via - <https://aws.amazon.com/ec2/pricing/>

Incorrect options:

Spot Instance - A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts (up to 90%), you can lower your Amazon EC2 costs significantly. Spot Instances are well-suited for data analysis, batch jobs, background processing, and other flexible tasks that can be interrupted. These can be terminated at short notice, so these are not suitable for critical workloads that need to run at a specific point in time.

Reserved Instance (RI) - Reserved Instances (RI) provide you with significant savings (up to 75%) on your Amazon EC2 costs compared to On-Demand Instance pricing. Reserved Instances (RI) are not physical instances, but rather a billing discount applied to the use of On-Demand Instances in your account. You can purchase a Reserved Instance (RI) for a one-year or three-year commitment, with the three-year commitment offering a bigger discount. Reserved instances (RI) cannot be interrupted. Reserved instances (RI) are not the right choice for short-term workloads.

Dedicated Host - Amazon EC2 Dedicated Hosts allow you to use your eligible software licenses from vendors such as Microsoft and Oracle on Amazon EC2 so that you get the flexibility and cost-effectiveness of using your licenses, but with the resiliency, simplicity, and elasticity of AWS. An Amazon EC2 Dedicated Host is a physical server fully dedicated for your use, so you can help address corporate compliance requirement. They're not cost-efficient compared to On-Demand instances. So this option is not correct.

Reference:

<https://aws.amazon.com/ec2/pricing/>

Domain

Billing and Pricing

Question 7Incorrect

Which of the following Amazon Simple Storage Service (Amazon S3) storage classes has NO constraint of a minimum storage duration charge for objects?

Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

Your answer is incorrect

Amazon S3 Glacier Flexible Retrieval

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

Correct answer

Amazon S3 Standard

Overall explanation

Correct option:

Amazon S3 Standard

Amazon S3 Standard offers high durability, availability, and performance object storage for frequently accessed data. Amazon S3 Standard offers low latency and high throughput performance. It is designed for durability of 99.999999999% of objects across multiple Availability Zones (AZ). Amazon S3 Standard has no constraint of a minimum storage duration for objects.

Please review this illustration for Amazon S3 Storage Classes retrieval fee. You don't need to memorize the actual numbers, just remember that Amazon S3 Standard and Amazon S3 Intelligent-Tiering do not charge any retrieval fee:

Performance across the S3 Storage Classes

	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier Instant Retrieval	S3 Glacier Flexible Retrieval	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128 KB	128 KB	128 KB	40 KB	40 KB
Minimum storage duration charge	N/A	N/A	30 days	30 days	90 days	90 days	180 days
Retrieval charge	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	milliseconds	milliseconds	milliseconds	milliseconds	minutes or hours	hours
Storage type	Object	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes	Yes

via - <https://aws.amazon.com/s3/storage-classes/>

Incorrect options:

Amazon S3 Glacier Flexible Retrieval - Amazon S3 Glacier Flexible Retrieval is a secure, durable, and extremely low-cost Amazon S3 storage class for data archiving and long-term backup. It is designed to deliver 99.999999999% durability, and provide comprehensive security and compliance capabilities that can help meet even

the most stringent regulatory requirements. Amazon S3 Glacier Flexible Retrieval mandates a minimum storage duration charge for 90 days.

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) - Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is for data that is accessed less frequently, but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZ), Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) stores data in a single Availability Zone (AZ). It is not suitable for data archival. Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) mandates a minimum storage duration charge for 30 days.

Amazon S3 Standard-Infrequent Access (S3 Standard-IA) - Amazon S3 Standard-Infrequent Access (S3 Standard-IA) is for data that is accessed less frequently, but requires rapid access when needed. Amazon S3 Standard-Infrequent Access (S3 Standard-IA) offers high durability, high throughput, and low latency of Amazon S3 Standard, with a low per GB storage price and per GB retrieval fee. This combination of low cost and high performance makes Amazon S3 Standard-Infrequent Access (S3 Standard-IA) ideal for long-term storage, backups, and as a data store for disaster recovery files. Amazon S3 Standard-Infrequent Access (S3 Standard-IA) mandates a minimum storage duration charge for 30 days.

Reference:

<https://aws.amazon.com/s3/storage-classes/>

Domain

Technology

Question 8Incorrect

Which AWS service can help you analyze your infrastructure to identify unattached or underutilized Amazon EBS Elastic Volumes?

AWS Config

Amazon CloudWatch

Correct answer

AWS Trusted Advisor

Your answer is incorrect

Amazon Inspector

Overall explanation

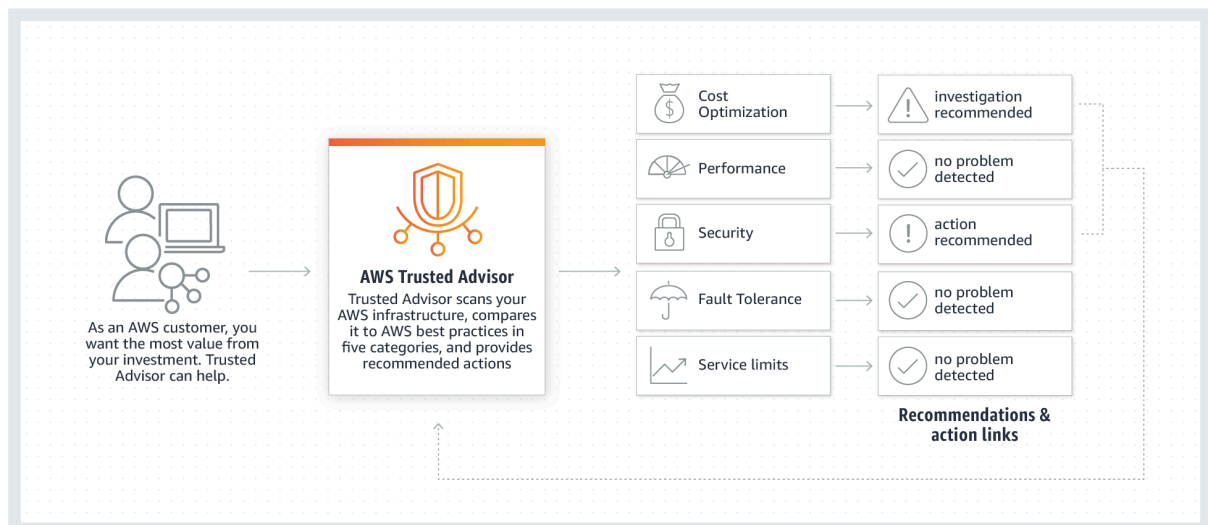
Correct option:

AWS Trusted Advisor

AWS Trusted Advisor is an online tool that provides real-time guidance to help provision your resources following AWS best practices. Whether establishing new workflows, developing applications, or as part of ongoing improvement, recommendations provided by Trusted Advisor regularly help keep your solutions provisioned optimally. AWS Trusted Advisor analyzes your AWS environment and provides best practice recommendations in five categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits.

AWS Trusted Advisor can check Amazon Elastic Block Store (Amazon EBS) volume configurations and warns when volumes appear to be underused. Charges begin when a volume is created. If a volume remains unattached or has very low write activity (excluding boot volumes) for a period of time, the volume is probably not being used.

How AWS Trusted Advisor Works:



via - <https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Incorrect options:

AWS Config - AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations. Think resource-specific change history, audit, and compliance; think Config. It's a configuration tracking service and not an infrastructure tracking service.

Amazon CloudWatch - Amazon CloudWatch is a monitoring and observability service built for DevOps engineers, developers, site reliability engineers (SREs), and IT managers. CloudWatch provides data and actionable insights to monitor applications, respond to system-wide performance changes, optimize resource utilization, and get a unified view of operational health. Amazon EBS emits notifications based on Amazon CloudWatch Events for a variety of volume, snapshot, and encryption status changes. With CloudWatch Events, you can establish rules that trigger programmatic actions in response to a change in volume, snapshot, or encryption key state (though not for underutilized volume usage).

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on your Amazon EC2 instances. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. Its a security assessment service and not an infrastructure tracking service.

References:

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/best-practice-checklist/>

[https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-cloud-watch-events.h
tml](https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-cloud-watch-events.html)

Domain

Billing and Pricing

Question 9Correct

As per the AWS Shared Responsibility Model, which of the following is a responsibility of AWS from a security and compliance point of view?

Service and Communications Protection

Your answer is correct

Patching networking infrastructure

Patching guest OS and applications

Identity and Access Management

Overall explanation

Correct option:

Patching networking infrastructure

According to the AWS Shared Responsibility Model, AWS is responsible for "Security of the Cloud". This includes protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services. Therefore, patching networking infrastructure is the responsibility of AWS.

Incorrect options:

Service and Communications Protection

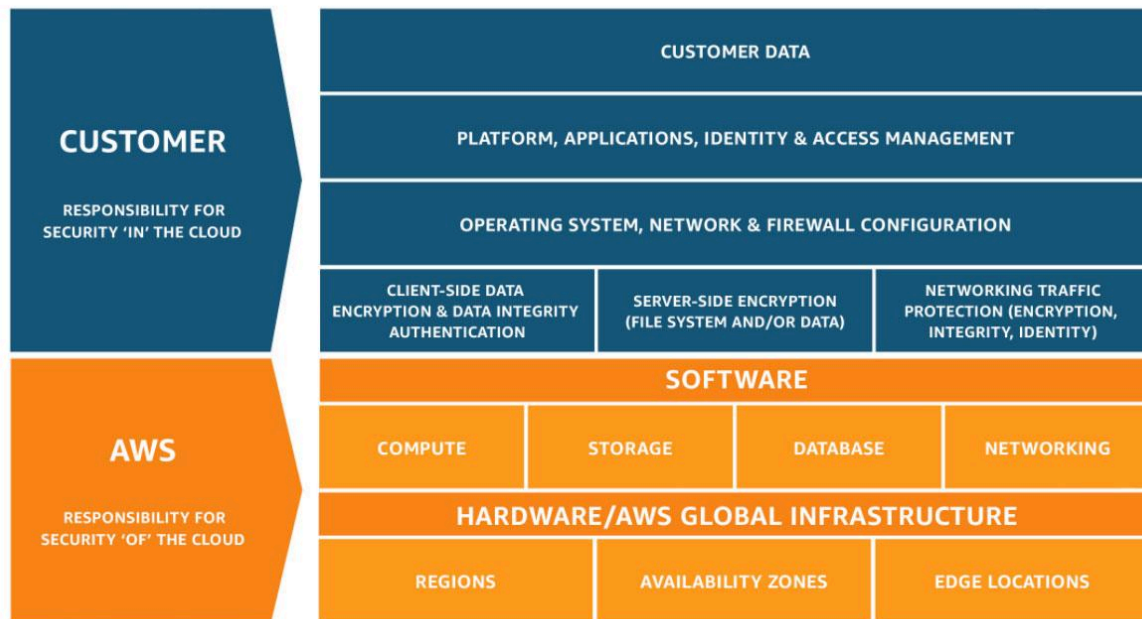
Identity and Access Management

Patching guest OS and applications

The customer is responsible for security "in" the cloud. This covers things such as services and communications protection; Identity and Access Management; and patching guest OS and applications. Customers are responsible for managing their data including encryption options and using Identity and Access Management tools for implementing appropriate access control policies as per their organization requirements. Therefore, these three options fall under the responsibility of the customer according to the AWS shared responsibility model.

Exam Alert:

Please review the AWS Shared Responsibility Model in detail as you can expect multiple questions on this topic in the exam:



via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Reference:

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Domain

Security and Compliance

Question 10Incorrect

Which of the following entities can be used to connect to an Amazon Elastic Compute Cloud (Amazon EC2) server from a Mac OS, Windows or Linux based computer via a browser-based client?

Correct answer

Amazon EC2 Instance Connect

Putty

Your answer is incorrect

AWS Direct Connect

SSH

Overall explanation

Correct option:

Amazon EC2 Instance Connect

Amazon EC2 Instance Connect provides a simple and secure way to connect to your instances using Secure Shell (SSH). With Amazon EC2 Instance Connect, you use AWS Identity and Access Management (AWS IAM) policies and principals to control

SSH access to your instances, removing the need to share and manage SSH keys. All connection requests using Amazon EC2 Instance Connect are logged to AWS CloudTrail so that you can audit connection requests.

You can use Amazon EC2 Instance Connect to connect to your Linux instances using a browser-based client, the Amazon EC2 Instance Connect CLI, or the SSH client of your choice. Amazon EC2 Instance Connect can be used to connect to an EC2 instance from a Mac OS, Windows or Linux based computer.

Incorrect options:

SSH - SSH can be used from a Mac OS, Windows or Linux based computer, but it's not a browser-based client.

Putty - Putty can be used only from Windows based computers.

AWS Direct Connect - AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. You can use AWS Direct Connect to establish a private virtual interface from your on-premise network directly to your Amazon VPC. This private connection takes at least one month for completion. AWS Direct Connect cannot be used to connect to an Amazon EC2 instance from a Mac OS, Windows or Linux based computer.

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Connect-using-EC2-Instance-Connect.html>

Domain

Cloud Concepts

Question 11 Incorrect

A financial services company wants to ensure that all customer data uploaded on its data lake on Amazon Simple Storage Service (Amazon S3) always stays private.

Which of the following is the MOST efficient solution to address this compliance requirement?

Set up a high-level advisory committee to review the privacy settings of each object uploaded into Amazon S3

Correct answer

Use Amazon S3 Block Public Access to ensure that all Amazon S3 resources stay private

Trigger a lambda function every time an object is uploaded on Amazon S3. The lambda function should change the object settings to make sure it stays private

Your answer is incorrect

Use Amazon CloudWatch to ensure that all Amazon S3 resources stay private

Overall explanation

Correct option:

Use Amazon S3 Block Public Access to ensure that all Amazon S3 resources stay private

The Amazon S3 Block Public Access feature provides settings for access points, buckets, and accounts to help you manage public access to Amazon S3 resources. By default, new buckets, access points, and objects don't allow public access. However, users can modify bucket policies, access point policies, or object permissions to allow public access. S3 Block Public Access settings override these policies and permissions so that you can limit public access to these resources.

When Amazon S3 receives a request to access a bucket or an object, it determines whether the bucket or the bucket owner's account has a block public access setting applied. If the request was made through an access point, Amazon S3 also checks for block public access settings for the access point. If there is an existing block public access setting that prohibits the requested access, Amazon S3 rejects the request.

Amazon S3 Block Public Access Overview:

Using Amazon S3 block public access

[PDF](#) | [Kindle](#) | [RSS](#)

The Amazon S3 Block Public Access feature provides settings for access points, buckets, and accounts to help you manage public access to Amazon S3 resources. By default, new buckets, access points, and objects don't allow public access. However, users can modify bucket policies, access point policies, or object permissions to allow public access. S3 Block Public Access settings override these policies and permissions so that you can limit public access to these resources.

With S3 Block Public Access, account administrators and bucket owners can easily set up centralized controls to limit public access to their Amazon S3 resources that are enforced regardless of how the resources are created.

When Amazon S3 receives a request to access a bucket or an object, it determines whether the bucket or the bucket owner's account has a block public access setting applied. If the request was made through an access point, Amazon S3 also checks for block public access settings for the access point. If there is an existing block public access setting that prohibits the requested access, Amazon S3 rejects the request.

Amazon S3 Block Public Access provides four settings. These settings are independent and can be used in any combination. Each setting can be applied to an access point, a bucket, or an entire AWS account. If the block public access settings for the access point, bucket, or account differ, then Amazon S3 applies the most restrictive combination of the access point, bucket, and account settings.

When Amazon S3 evaluates whether an operation is prohibited by a block public access setting, it rejects any request that violates an access point, bucket, or account setting.

Warning

Public access is granted to buckets and objects through access control lists (ACLs), access point policies, bucket policies, or all. To help ensure that all of your Amazon S3 access points, buckets, and objects have their public access blocked, we recommend that you turn on all four settings for block public access for your account. These settings block public access for all current and future buckets and access points.

Before applying these settings, verify that your applications will work correctly without public access. If you require some level of public access to your buckets or objects, for example to host a static website as described at [Hosting a static website on Amazon S3](#), you can customize the individual settings to suit your storage use cases.

via -

<https://docs.aws.amazon.com/AmazonS3/latest/dev/access-control-block-public-access.html>

Incorrect options:

Trigger a lambda function every time an object is uploaded on Amazon S3. The lambda function should change the object settings to make sure it stays private -
Although it's possible to implement this solution, but it is more efficient to use the "Amazon S3 Block Public Access" feature as its available off-the-shelf.

Use Amazon CloudWatch to ensure that all Amazon S3 resources stay private -

Amazon CloudWatch is a monitoring and observability service built for DevOps engineers, developers, site reliability engineers (SREs), and IT managers.

CloudWatch provides data and actionable insights to monitor applications, respond

to system-wide performance changes, optimize resource utilization, and get a unified view of operational health. This is an excellent service for building Resilient systems. Think resource performance monitoring, events, and alerts; think CloudWatch. CloudWatch cannot be used to ensure data privacy on Amazon S3.

Set up a high-level advisory committee to review the privacy settings of each object uploaded into Amazon S3 - This option has been added as a distractor.

Reference:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/access-control-block-public-access.html>

Domain

Security and Compliance

Question 12Incorrect

Which benefit of Cloud Computing allows AWS to offer lower pay-as-you-go prices as usage from hundreds of thousands of customers is aggregated in the cloud?

Correct answer

Massive economies of scale

Trade capital expense for variable expense

Increased speed and agility

Your answer is incorrect

Go global in minutes

Overall explanation

Correct option:

Massive economies of scale

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis.

By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay-as-you-go prices.

Exam Alert:

Please check out the following six advantages of Cloud Computing. You would certainly be asked questions on the advantages of Cloud Computing compared to a traditional on-premises setup:

Six Advantages of Cloud Computing

PDF  | [RSS](#)

- **Trade capital expense for variable expense** – Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.
- **Benefit from massive economies of scale** – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.
- **Stop guessing capacity** – Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision prior to deploying an application, you often end up either sitting on expensive idle resources or dealing with limited capacity. With cloud computing, these problems go away. You can access as much or as little capacity as you need, and scale up and down as required with only a few minutes' notice.
- **Increase speed and agility** – In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.
- **Stop spending money running and maintaining data centers** – Focus on projects that differentiate your business, not the infrastructure. Cloud computing lets you focus on your own customers, rather than on the heavy lifting of racking, stacking, and powering servers.
- **Go global in minutes** – Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

via -

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Incorrect options:

Trade capital expense for variable expense - Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.

Increased speed and agility - In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization since the cost and time it takes to experiment and develop is significantly lower.

Go global in minutes - Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

Although these three options are also benefits of Cloud Computing, it is the massive economies of scale that allow AWS to offer lower pay-as-you-go prices as usage from hundreds of thousands of customers is aggregated in the cloud.

References:

<https://aws.amazon.com/what-is-cloud-computing/>

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Domain

Cloud Concepts

Question 13Correct

A social media company wants to have the MOST cost-optimal strategy for deploying Amazon Elastic Compute Cloud (Amazon EC2) instances. As a Cloud Practitioner, which of the following options would you recommend? (Select two)

Your selection is correct

Use Spot Instances for ad-hoc jobs that can be interrupted

Use Reserved Instances (RI) for ad-hoc jobs that can be interrupted

Use On-Demand Instances for ad-hoc jobs that can be interrupted

Use On-Demand Instances to run applications with a predictable usage over the next one year

Your selection is correct

Use Reserved Instances (RI) to run applications with a predictable usage over the next one year

Overall explanation

Correct options:

Use Spot Instances for ad-hoc jobs that can be interrupted

A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts (up to 90%), you can lower your Amazon EC2 costs significantly. Spot Instances are well-suited for data analysis, batch jobs, background processing, and optional tasks. These can be terminated at short notice, so these are not suitable for critical workloads that need to run at a specific point in time.

Use Reserved Instances (RI) to run applications with a predictable usage over the next one year

Reserved Instances (RI) provide you with significant savings (up to 75%) on your Amazon EC2 costs compared to On-Demand Instance pricing. Reserved Instances (RI) are not physical instances, but rather a billing discount applied to the use of

On-Demand Instances in your account. You can purchase a Reserved Instance (RI) for a one-year or three-year commitment, with the three-year commitment offering a bigger discount. Reserved instances (RI) are a great fit for application with a steady-state usage. Reserved instances (RI) cannot be interrupted.

Amazon EC2 Pricing Options Overview:

On-Demand

With On-Demand instances, you pay for compute capacity by the hour or the second depending on which instances you run. No longer-term commitments or upfront payments are needed. You can increase or decrease your compute capacity depending on the demands of your application and only pay the specified per hourly rates for the instance you use.

On-Demand instances are recommended for:

- Users that prefer the low cost and flexibility of Amazon EC2 without any up-front payment or long-term commitment
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

[See On-Demand pricing »](#)

Spot instances

Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. [Learn More](#).

Spot instances are recommended for:

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with urgent computing needs for large amounts of additional capacity

[See Spot pricing »](#)

Savings Plans

Savings Plans are a flexible pricing model that offer low prices on EC2 and Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term.

Dedicated Hosts

A Dedicated Host is a physical EC2 server dedicated for your use. Dedicated Hosts can help you reduce costs by allowing you to use your existing server-bound software licenses, including Windows Server, SQL Server, and SUSE Linux Enterprise Server (subject to your license terms), and can also help you meet compliance requirements. [Learn more](#).

- Can be purchased On-Demand (hourly).
- Can be purchased as a Reservation for up to 70% off the On-Demand price.

[See Dedicated pricing »](#)

Reserved Instances

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them.

For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. See [How to Purchase Reserved Instances](#) for more information.

Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

via - <https://aws.amazon.com/ec2/pricing/>

Incorrect options:

Use On-Demand Instances to run applications with a predictable usage over the next one year

Use On-Demand Instances for ad-hoc jobs that can be interrupted

An On-Demand Instance is an instance that you use on-demand. You have full control over its lifecycle — you decide when to launch, stop, hibernate, start, reboot, or terminate it. There is no long-term commitment required when you purchase On-Demand Instances. There is no upfront payment and you pay only for the seconds that your On-Demand Instances are running. The price per second for running an On-Demand Instance is fixed. On-demand instances cannot be

interrupted. However, On-demand instances are not as cost-effective as Spot instances or Reserved instances (RI), so both these options are not correct.

Use Reserved Instances (RI) for ad-hoc jobs that can be interrupted - Spot instances are more cost-effective than Reserved instances (RI) for running ad-hoc jobs that can be interrupted, so this option is not correct.

Reference:

<https://aws.amazon.com/ec2/pricing/>

Domain

Billing and Pricing

Question 14Correct

Which AWS services can be used together to send alerts whenever the AWS account root user signs in? (Select two)

Your selection is correct

Amazon CloudWatch

Your selection is correct

Amazon Simple Notification Service (Amazon SNS)

AWS Lambda

AWS Step Functions

Amazon Simple Queue Service (Amazon SQS)

Overall explanation

Correct options:

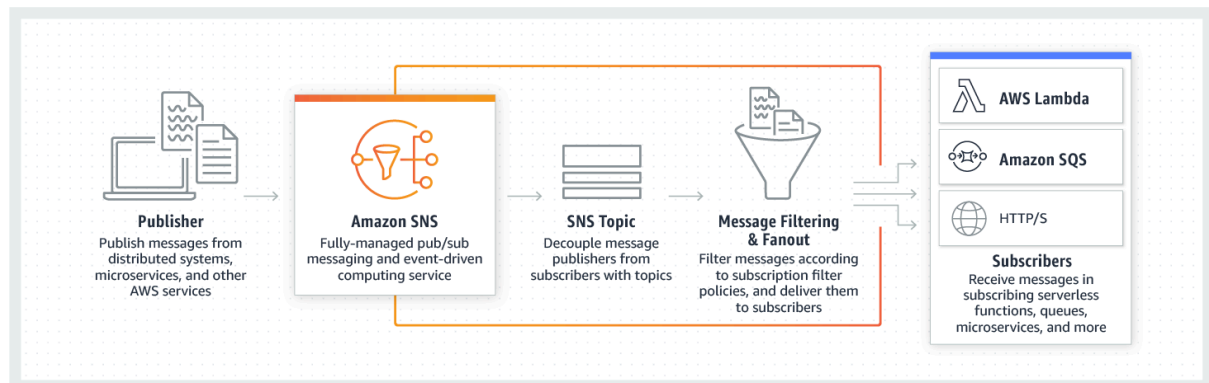
Amazon Simple Notification Service (Amazon SNS)

Amazon CloudWatch

Amazon CloudWatch Events delivers a near real-time stream of system events that describe changes in Amazon Web Services (AWS) resources. Using simple rules that you can quickly set up, you can match events and route them to one or more target functions or streams. Amazon CloudWatch Events becomes aware of operational changes as they occur. Amazon CloudWatch Events responds to these operational changes and takes corrective action as necessary, by sending messages to respond to the environment, activating functions, making changes, and capturing state information.

Amazon Simple Notification Service (Amazon SNS) is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and serverless applications. Additionally, Amazon SNS can be used to fan out notifications to end users using mobile push, SMS, and email.

How Amazon SNS Works:



via - <https://aws.amazon.com/sns/>

To send alerts whenever the AWS account root user signs in, you can create an Amazon Simple Notification Service (Amazon SNS) topic. Then, create an Amazon CloudWatch event rule to monitor userIdentity root logins from the AWS Management Console and send an email via Amazon SNS when the event triggers.

Incorrect options:

Amazon Simple Queue Service (Amazon SQS) - Amazon Simple Queue Service (Amazon SQS) is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications. Using Amazon SQS, you can send, store, and receive messages between software components at any volume, without losing messages or requiring other services to be available.

AWS Lambda - AWS Lambda is a compute service that lets you run code without provisioning or managing servers.

AWS Step Functions - AWS Step Functions lets you coordinate multiple AWS services into serverless workflows. You can design and run workflows that stitch together services such as AWS Lambda, AWS Glue and Amazon SageMaker.

References:

<https://aws.amazon.com/premiumsupport/knowledge-center/root-user-account-cloudwatch-rule/>

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/events/WhatIsCloudWatchEvents.html>

Domain

Security and Compliance

Question 15Incorrect

AWS Shield Advanced provides expanded DDoS attack protection for web applications running on which of the following resources? (Select two)

Correct selection

Amazon CloudFront

Your selection is incorrect

AWS Identity and Access Management (AWS IAM)

Your selection is incorrect

Amazon Simple Storage Service (Amazon S3)

AWS Elastic Beanstalk

Correct selection

Amazon Elastic Compute Cloud (Amazon EC2)

Overall explanation

Correct options:

Amazon CloudFront

Amazon Elastic Compute Cloud (Amazon EC2)

AWS Shield Standard is activated for all AWS customers, by default. For higher levels of protection against attacks, you can subscribe to AWS Shield Advanced. With Shield Advanced, you also have exclusive access to advanced, real-time metrics and reports for extensive visibility into attacks on your AWS resources. With the assistance of the DRT (DDoS response team), AWS Shield Advanced includes intelligent DDoS attack detection and mitigation for not only for network layer (layer 3) and transport layer (layer 4) attacks but also for application layer (layer 7) attacks.

AWS Shield Advanced provides expanded DDoS attack protection for web applications running on the following resources: Amazon Elastic Compute Cloud,

Elastic Load Balancing (ELB), Amazon CloudFront, Amazon Route 53, AWS Global Accelerator.

Incorrect options:

Amazon Simple Storage Service (Amazon S3)

AWS Elastic Beanstalk

AWS Identity and Access Management (AWS IAM)

These three resource types are not supported by AWS Shield Advanced.

Reference:

<https://docs.aws.amazon.com/waf/latest/developerguide/ddos-overview.html>

Domain

Security and Compliance

Question 16Correct

The QA team at a company wants a tool/service that can provide access to different mobile devices with variations in firmware and Operating System versions.

Which AWS service can address this use case?

Your answer is correct

AWS Device Farm

AWS Elastic Beanstalk

AWS CodePipeline

AWS Mobile Farm

Overall explanation

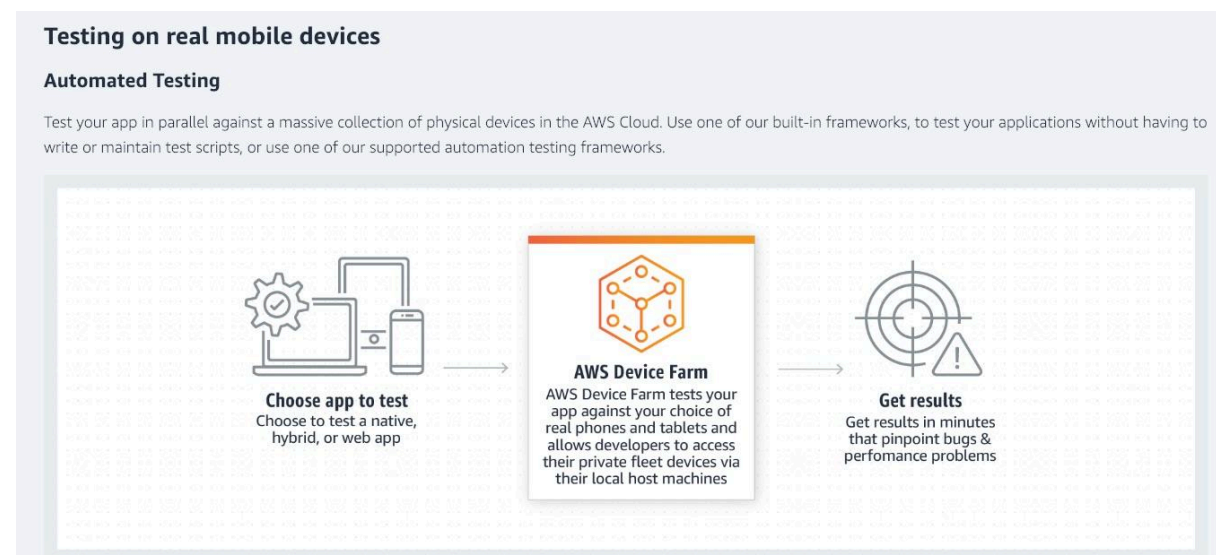
Correct option:

AWS Device Farm

AWS Device Farm is an application testing service that lets you improve the quality of your web and mobile apps by testing them across an extensive range of desktop browsers and real mobile devices; without having to provision and manage any testing infrastructure. The service enables you to run your tests concurrently on multiple desktop browsers or real devices to speed up the execution of your test suite, and generates videos and logs to help you quickly identify issues with your app.

Device Farm is designed for developers, QA teams, and customer support representatives who are building, testing, and supporting mobile apps to increase the quality of their apps. Application quality is increasingly important, and also getting complex due to the number of device models, variations in firmware and OS versions, carrier and manufacturer customizations, and dependencies on remote services and other apps. AWS Device Farm accelerates the development process by executing tests on multiple devices, giving developers, QA and support professionals the ability to perform automated tests and manual tasks like reproducing customer issues, exploratory testing of new functionality, and executing manual test plans. AWS Device Farm also offers significant savings by eliminating the need for internal device labs, lab managers, and automation infrastructure development.

How AWS Device Farm works:



via - <https://aws.amazon.com/device-farm/>

Incorrect options:

AWS CodePipeline - AWS CodePipeline is a fully managed continuous delivery service that helps you automate your release pipelines for fast and reliable application and infrastructure updates. CodePipeline automates the build, test, and deploy phases of your release process every time there is a code change, based on the release model you define. This enables you to rapidly and reliably deliver features and updates.

AWS Elastic Beanstalk - AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, etc.

You can simply upload your code and Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring. At the same time, you retain full control over the AWS resources powering your application and can access the underlying resources at any time.

AWS Mobile Farm - This is an invalid option, given only as a distractor.

Reference:

<https://aws.amazon.com/device-farm/>

Domain

Technology

Question 17Incorrect

AWS Organizations provides which of the following benefits? (Select two)

Correct selection

Share the reserved Amazon EC2 instances amongst the member AWS accounts

Your selection is incorrect

Check vulnerabilities on Amazon EC2 instances across the member AWS accounts

Provision Amazon EC2 Spot instances across the member AWS accounts

Your selection is incorrect

Deploy patches on Amazon EC2 instances across the member AWS accounts

Correct selection

Volume discounts for Amazon EC2 and Amazon S3 aggregated across the member AWS accounts

Overall explanation

Correct options:

Volume discounts for Amazon EC2 and Amazon S3 aggregated across the member AWS accounts

Share the reserved Amazon EC2 instances amongst the member AWS accounts

AWS Organizations helps you to centrally manage billing; control access, compliance, and security; and share resources such as reserved Amazon EC2 instances across your AWS accounts.

Using AWS Organizations, you can automate account creation, create groups of accounts to reflect your business needs, and apply policies for these groups for governance. You can also simplify billing by setting up a single payment method for all of your AWS accounts. AWS Organizations is available to all AWS customers at no additional charge.

You can use AWS Organizations to set up a single payment method for all the AWS accounts in your organization through consolidated billing. With consolidated billing, you can see a combined view of charges incurred by all your accounts, as well as take advantage of pricing benefits from aggregated usage, such as volume discounts for Amazon EC2 and Amazon S3.

Key benefits of AWS Organizations:

CENTRALLY MANAGE POLICIES ACROSS MULTIPLE AWS ACCOUNTS

To improve control over your AWS environment, you can use AWS Organizations to create groups of accounts, and then attach policies to a group to ensure the correct policies are applied across the accounts without requiring custom scripts and manual processes.

AUTOMATE AWS ACCOUNT CREATION AND MANAGEMENT

AWS Organizations helps you simplify IT operations by automating AWS account creation and management. The Organizations APIs enable you to create new accounts programmatically, and to add the new accounts to a group. The policies attached to the group are automatically applied to the new account. For example, you can automate the creation of new accounts for workload or application isolation and grant entities in those accounts access only to the necessary AWS services.

CONSOLIDATE BILLING ACROSS MULTIPLE AWS ACCOUNTS

You can use AWS Organizations to set up a single payment method for all the AWS accounts in your organization through consolidated billing. With consolidated billing, you can see a combined view of charges incurred by all your accounts, as well as take advantage of pricing benefits from aggregated usage, such as volume discounts for Amazon EC2 and Amazon S3.

GOVERN ACCESS TO AWS SERVICES, RESOURCES, AND REGIONS

AWS Organizations allows you to restrict what services and actions are allowed in your accounts. You can use Service Control Policies (SCPs) to apply permission guardrails on [AWS Identity and Access Management \(IAM\)](#) users and roles. For example, you can apply an SCP that restricts users in accounts in your organization from launching any resources in regions that you do not explicitly allow.

CONFIGURE AWS SERVICES ACROSS MULTIPLE ACCOUNTS

AWS Organizations helps you configure [AWS services](#) and share resources across accounts in your organization. For example, Organizations integrates with [AWS Single Sign-on](#) to enable you to easily provision access for all of your developers to accounts in your organization from a single place. You can make central changes to access permissions and have them automatically updated on accounts in your organization.

via - <https://aws.amazon.com/organizations/>

Incorrect options:

Check vulnerabilities on Amazon EC2 instances across the member AWS accounts

Deploy patches on Amazon EC2 instances across the member AWS accounts

Provision Amazon EC2 Spot instances across the member AWS accounts

These three options contradict the details provided earlier in the explanation, so these options are incorrect.

Reference:

<https://aws.amazon.com/organizations/>

Domain

Billing and Pricing

Question 18Correct

An e-commerce company would like to receive alerts when the Amazon EC2 Reserved Instances (RI) utilization drops below a certain threshold. Which AWS service can be used to address this use-case?

AWS Cost Explorer

AWS Trusted Advisor

Your answer is correct

AWS Budgets

AWS Systems Manager

Overall explanation

Correct option:

AWS Budgets

AWS Budgets gives you the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. You can define a utilization threshold and receive alerts when your Reserved Instances (RI) usage falls below that threshold. This lets you see if your Reserved Instances (RI) are unused or under-utilized. Reservation alerts are supported for Amazon EC2, Amazon RDS, Amazon Redshift, Amazon ElastiCache, and Amazon Elasticsearch reservations.

AWS Budgets Overview:

Managing your costs with AWS Budgets

[PDF](#) | [Kindle](#) | [RSS](#)

AWS Budgets enable you to plan your service usage, service costs, and instance reservations. Budgets provide you with a way to see the following information:

- How close your plan is to your budgeted amount or to the free tier limits
- Your usage to date, including how much you have used of your Reserved Instances (RIs)
- Your current estimated charges from AWS and how much your predicted usage will incur in charges by the end of the month
- How much of your budget has been used

AWS Budgets information is updated up to three times a day. Updates typically occur between 8 to 12 hours after the previous update. Budgets track your unblended costs, subscriptions, refunds, and RIs. You can create the following types of budgets:

- **Cost budgets** – Plan how much you want to spend on a service.
- **Usage budgets** – Plan how much you want to use one or more services.
- **RI utilization budgets** – Define a utilization threshold and receive alerts when your RI usage falls below that threshold. This lets you see if your RIs are unused or under-utilized.
- **RI coverage budgets** – Define a coverage threshold and receive alerts when the number of your instance hours that are covered by RIs fall below that threshold. This lets you see how much of your instance usage is covered by a reservation.
- **Savings Plans utilization budgets** – Define a utilization threshold and receive alerts when the usage of your Savings Plans falls below that threshold. This lets you see if your Savings Plans are unused or under-utilized.
- **Savings Plans coverage budgets** – Define a coverage threshold and receive alerts when your Savings Plans eligible usage that is covered by Savings Plans fall below that threshold. This lets you see how much of your instance usage is covered by Savings Plans.

via -

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/budgets-managing-costs.html>

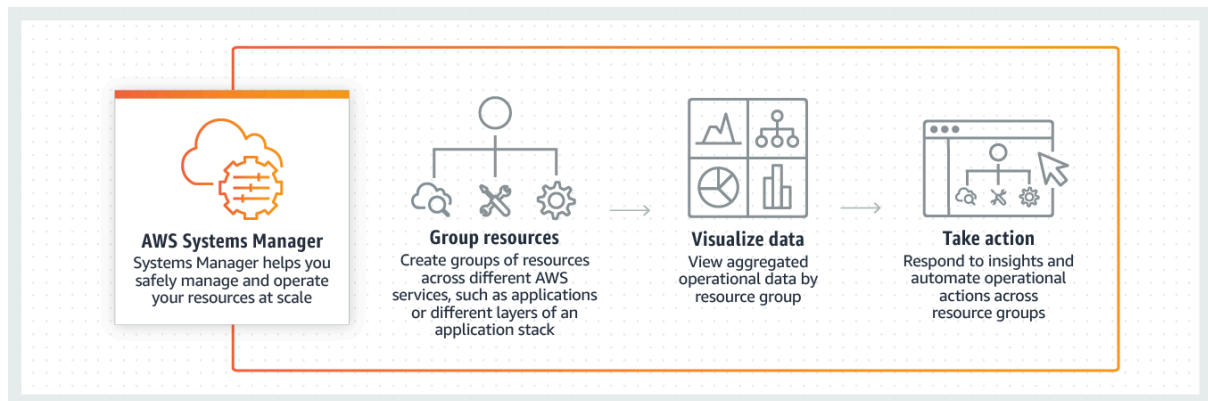
Incorrect options:

AWS Trusted Advisor - AWS Trusted Advisor is an online tool that provides real-time guidance to help provision your resources following AWS best practices. Whether establishing new workflows, developing applications, or as part of ongoing improvement, recommendations provided by AWS Trusted Advisor regularly help keep your solutions provisioned optimally. AWS Trusted Advisor analyzes your AWS environment and provides best practice recommendations in five categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits.

AWS Cost Explorer - AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown of all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends. You can use AWS Cost Explorer to generate the RI utilization and RI

coverage reports, but you cannot use it to generate alerts for under-utilized Reserved Instances (RI).

AWS Systems Manager - AWS Systems Manager gives you visibility and control of your infrastructure on AWS. AWS Systems Manager provides a unified user interface so you can view operational data from multiple AWS services and allows you to automate operational tasks such as running commands, managing patches, and configuring servers across AWS Cloud as well as on-premises infrastructure.



via - <https://aws.amazon.com/systems-manager/>

Reference:

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/budgets-managing-costs.html>

Domain

Billing and Pricing

Question 19Incorrect

Which of the following are benefits of the AWS Web Application Firewall (AWS WAF)?
(Select two)

Your selection is incorrect

AWS Web Application Firewall (AWS WAF) offers protection against all known infrastructure (Layer 3 and 4) attacks

Correct selection

AWS Web Application Firewall (AWS WAF) can check for the presence of SQL code that is likely to be malicious (known as SQL injection)

Your selection is correct

AWS Web Application Firewall (AWS WAF) can block all requests except the ones that you allow

AWS Web Application Firewall (AWS WAF) offers dedicated support from the DDoS Response Team (DRT) and advanced reporting

AWS Web Application Firewall (AWS WAF) lets you monitor the HTTP and HTTPS requests that are forwarded to Amazon Route 53

Overall explanation

Correct options:

AWS Web Application Firewall (AWS WAF) can block all requests except the ones that you allow

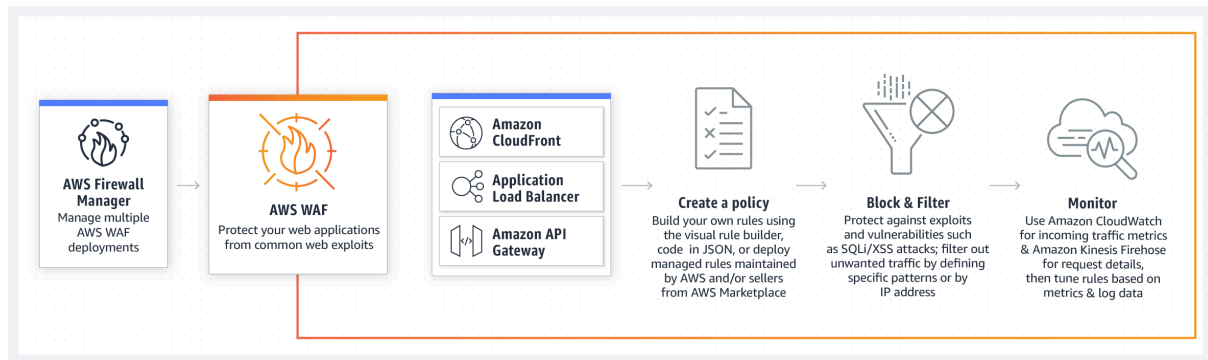
AWS Web Application Firewall (AWS WAF) can block all requests except the ones that you allow. This is useful when you want to serve content for a restricted website whose users are readily identifiable by properties in web requests, such as the IP addresses that they use to browse to the website.

AWS Web Application Firewall (AWS WAF) can check for the presence of SQL code that is likely to be malicious (known as SQL injection)

AWS Web Application Firewall (AWS WAF) offers additional protection against web attacks using conditions that you specify. You can define conditions by using characteristics of web requests such as - IP addresses that requests originate from, presence of a script that is likely to be malicious (known as cross-site scripting), presence of SQL code that is likely to be malicious (known as SQL injection) and many more.

AWS Web Application Firewall (AWS WAF) is a web application firewall that helps protect your web applications or APIs against common web exploits that may affect availability, compromise security, or consume excessive resources. AWS Web Application Firewall (AWS WAF) gives you control over how traffic reaches your applications by enabling you to create security rules that block common attack patterns such as SQL injection or cross-site scripting. You can also use rate-based rules to mitigate the Web layer DDoS attack.

How AWS Web Application Firewall (AWS WAF) Works:



via - <https://aws.amazon.com/waf/>

Incorrect options:

AWS Web Application Firewall (AWS WAF) offers protection against all known infrastructure (Layer 3 and 4) attacks - AWS Web Application Firewall (AWS WAF) lets you monitor the HTTP and HTTPS requests to your application, it only works at the application layer (layer 7).

AWS Web Application Firewall (AWS WAF) offers dedicated support from the DDoS Response Team (DRT) and advanced reporting - As AWS Shield Advanced customer can contact a 24x7 DDoS response team (DRT) for assistance during a DDoS attack, it is a feature of Shield Advanced, and not of AWS Web Application Firewall (AWS WAF).

AWS Web Application Firewall (AWS WAF) lets you monitor the HTTP and HTTPS requests that are forwarded to Amazon Route 53 - AWS Web Application Firewall (AWS WAF) is a web application firewall that lets you monitor the HTTP and HTTPS requests that are forwarded to an Amazon API Gateway API, Amazon CloudFront or an Application Load Balancer. It does not cover Amazon Route 53, which is a Domain Name System (DNS) web service.

Reference:

<https://docs.aws.amazon.com/waf/latest/developerguide/what-is-aws-waf.html>

Domain

Security and Compliance

Question 20Incorrect

Which of the following AWS storage services can be directly used with on-premises systems?

Correct answer

Amazon Elastic File System (Amazon EFS)

Amazon EC2 Instance Store

Amazon Simple Storage Service (Amazon S3)

Your answer is incorrect

Amazon Elastic Block Store (Amazon EBS)

Overall explanation

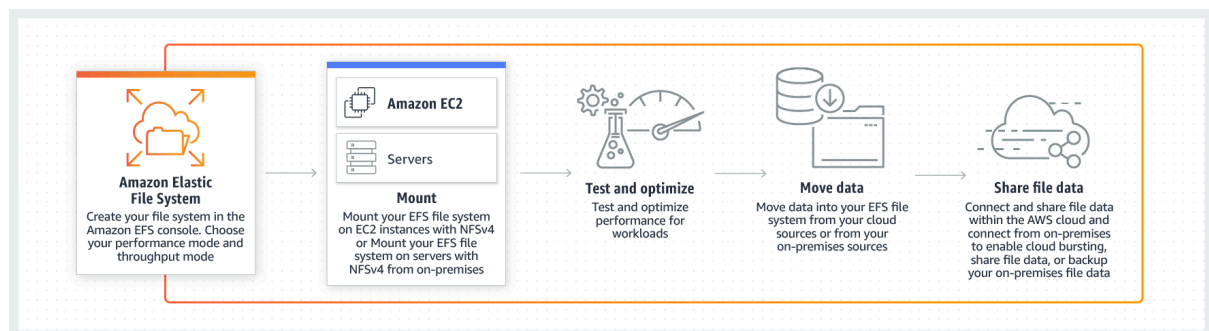
Correct option:

Amazon Elastic File System (Amazon EFS)

Amazon Elastic File System (Amazon EFS) provides a simple, scalable, fully managed elastic NFS file system for use with AWS Cloud services and on-premises resources.

To access EFS file systems from on-premises, you must have an AWS Direct Connect or AWS VPN connection between your on-premises datacenter and your Amazon VPC. You mount an EFS file system on your on-premises Linux server using the standard Linux mount command for mounting a file system

How Amazon EFS Works:



via - <https://aws.amazon.com/efs/faq/>

Incorrect options:

Amazon Elastic Block Store (Amazon EBS) - Amazon Elastic Block Store (EBS) is an easy to use, high-performance block storage service designed for use with Amazon Elastic Compute Cloud (EC2) for both throughput and transaction-intensive workloads at any scale. EBS volumes can only be mounted with Amazon EC2.

Amazon EC2 Instance Store - An instance store provides temporary block-level storage for your Amazon EC2 instance. This storage is located on disks that are

physically attached to the host computer. It is not possible to use this storage from on-premises systems.

Amazon Simple Storage Service (Amazon S3) - Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Amazon S3 can be accessed from on-premises only via AWS Storage Gateway. It is not possible to access S3 directly from on-premises systems.

Reference:

<https://aws.amazon.com/efs/faq/>

Domain

Cloud Concepts

Question 21 Correct

Which of the following AWS services can be used to forecast your AWS account usage and costs?

AWS Budgets

AWS Cost & Usage Report (AWS CUR)

AWS Pricing Calculator

Your answer is correct

AWS Cost Explorer

Overall explanation

Correct option:

AWS Cost Explorer

AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown of all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends. AWS Cost Explorer also supports forecasting to get a better idea of what your costs and usage may look like in the future so that you can plan.

AWS Cost Explorer Features:

AWS Cost Explorer Features

Get started quickly A set of default reports are included to help you quickly gain insight into your cost drivers and usage trends.	Set time interval and granularity Set a custom time period, and determine whether you would like to view your data at a monthly or daily level of granularity.	Filter/Group your data Dig deeper into your data by taking advantage of filtering and grouping functionality, using a variety of available dimensions.
Forecast future costs and usage Use forecasting to get a better idea of your what your costs and usage may look like in the future, so that you can plan ahead.	Save your progress Once you arrive at a helpful view, save your progress as a new report that you can refer back to in the future.	Build custom applications Directly access the interactive, ad-hoc analytics engine that powers AWS Cost Explorer.

via - <https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

Incorrect options:

AWS Cost & Usage Report (AWS CUR) - The AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can use Cost and Usage Reports to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself. AWS updates the report in your bucket once a day in a comma-separated value (CSV) format. AWS Cost and Usage Reports cannot forecast your AWS account cost and usage.

AWS Budgets - AWS Budgets gives the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. Budgets can be created at the monthly, quarterly, or yearly level, and you can customize the start and end dates. You can further refine your budget to track costs associated with multiple dimensions, such as AWS service, linked account, tag, and others. AWS Budgets cannot forecast your AWS account cost and usage.

AWS Pricing Calculator - AWS Pricing Calculator lets you explore AWS services and create an estimate for the cost of your use cases on AWS. You can model your solutions before building them, explore the price points and calculations behind your estimate, and find the available instance types and contract terms that meet your needs. This enables you to make informed decisions about using AWS. You can plan your AWS costs and usage or price out setting up a new set of instances and services. You cannot use this service to forecast your AWS account cost and usage.

Reference:

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

Domain

Billing and Pricing

Question 22Correct

Which of the following is available across all AWS Support plans?

Your answer is correct

AWS Health Dashboard – Your account health

Third-Party Software Support

Enhanced Technical Support with unlimited cases and unlimited contacts

Full set of AWS Trusted Advisor best practice checks

Overall explanation

Correct option:

AWS Health Dashboard – Your account health

Full set of AWS Trusted Advisor best practice checks, enhanced Technical Support with unlimited cases, and unlimited contacts and third-party Software Support are available only for Business and Enterprise Support plans.

AWS Health Dashboard – Your account health is available for all Support plans.

AWS Health Dashboard – Your account health provides alerts and remediation guidance when AWS is experiencing events that may impact you.

With AWS Health Dashboard – Your account health, alerts are triggered by changes in the health of your AWS resources, giving you event visibility, and guidance to help quickly diagnose and resolve issues.

You can check on this page <https://health.aws.amazon.com/health/home> to get current status information.

Exam Alert:

Please review the differences between the AWS Developer Support, AWS Business Support, AWS Enterprise On-Ramp Support and AWS Enterprise Support plans as you can expect at least a couple of questions on the exam:

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
AWS Trusted Advisor Best Practice Checks	Service Quota and basic Security checks	Full set of checks	Full set of checks	Full set of checks
AWS Trusted Advisor Priority				Prioritized recommendations curated by your AWS account team
Enhanced Technical Support	Business hours** web access to Cloud Support Associates Unlimited cases with 1 primary contact Prioritized responses on AWS re:Post	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack
Case Severity / Response Times*	General guidance: < 24 hours** System impaired: < 12 hours**	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business-critical system down: < 30 minutes	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business/Mission-critical system down: < 15 minutes
Architectural Guidance	General	Contextual to your use-cases	Consultative review and guidance based on your applications (one-per-year)	Consultative review and guidance based on your applications
Programmatic Case Management		AWS Support API	AWS Support API	AWS Support API

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Third-Party Software Support		Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting
Proactive Programs and Self Service	Access to Support Automation Workflows with prefixes AWSsupport	Access to Infrastructure Event Management for additional fee Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Infrastructure Event Management (one-per-year) Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Access to Infrastructure Event Management Access to proactive reviews, workshops, and deep dives Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport
AWS Incident Detection and Response				Access to AWS Incident Detection and Response for an additional fee. AWS Incident Detection and Response is an add-on to Enterprise Support that offers 24x7 proactive monitoring and incident management for selected workloads. AWS Incident Detection and Response leverages the proven operational, enhanced monitoring, and incident management capabilities used internally by AWS teams and externally by AWS Managed Services (AMS).
AWS Managed Services		Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud operations skills and capacity. It includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team. AWS Incident Detection and Response is available at no additional charge in eligible regions for AWS Managed Services direct customers with AWS Enterprise Support.

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Technical Account Management			A pool of Technical Account Managers to provide proactive guidance, and coordinate access to programs and AWS experts	Designated Technical Account Manager (TAM) to proactively monitor your environment and assist with optimization and coordinate access to programs and AWS experts
Training				Access to online self-paced labs
Account Assistance			Concierge Support Team	Concierge Support Team
Pricing	Greater of \$29 / month*** - or - 3% of monthly AWS usage See pricing detail and example.	Greater of \$100 / month*** - or - 10% of monthly AWS usage for the first \$0–\$10K 7% of monthly AWS usage from \$10K–\$80K 5% of monthly AWS usage from \$80K–\$250K 3% of monthly AWS usage over \$250K See pricing detail and example.	Greater of \$5,500 - or - 10% of monthly AWS usage See pricing detail and example.	Greater of \$15,000 - or - 10% of monthly AWS usage for the first \$0–\$150K 7% of monthly AWS usage from \$150K–\$500K 5% of monthly AWS usage from \$500K–\$1M 3% of monthly AWS usage over \$1M See pricing detail and example.
*Additional services for additional fee		*Access to AWS Managed Services (AMS) for an additional fee	*Access to AWS Managed Services (AMS) for an additional fee	* Access to AWS Incident Detection and Response for an additional fee. *Access to AWS Managed Services (AMS) for an additional fee

via - <https://aws.amazon.com/premiumsupport/plans/>

Incorrect options:

Full set of AWS Trusted Advisor best practice checks

Enhanced Technical Support with unlimited cases and unlimited contacts

Third-Party Software Support

As mentioned in the explanation above, these options are available only for AWS Business Support, AWS Enterprise On-Ramp Support and AWS Enterprise Support plans.

Reference:

<https://aws.amazon.com/premiumsupport/plans/>

Domain

Billing and Pricing

Question 23Incorrect

A cargo shipping company runs its server fleet on Amazon Elastic Compute Cloud (Amazon EC2) instances. Some of these instances host the CRM (Customer Relationship Management) applications that need to be accessible 24*7. These applications are not mission-critical. In case of a disaster, these applications can be managed on a lesser number of instances for some time.

Which disaster recovery strategy is well-suited as well as cost-effective for this requirement?

Backup & Restore strategy

Pilot Light strategy

Correct answer

Warm Standby strategy

Your answer is incorrect

Multi-site active-active strategy

Overall explanation

Correct option:

Warm Standby strategy

When selecting your disaster recovery (DR) strategy, you must weigh the benefits of lower RTO (recovery time objective) and RPO (recovery point objective) vs the costs of implementing and operating a strategy. The pilot light and warm standby strategies both offer a good balance of benefits and cost.

This strategy replicates data from the primary Region to data resources in the recovery Region, such as Amazon Relational Database Service (Amazon RDS) DB instances or Amazon DynamoDB tables. These data resources are ready to serve requests. In addition to replication, this strategy requires you to create a continuous backup in the recovery Region. This is because when "human action" type disasters occur, data can be deleted or corrupted, and replication will replicate the bad data. Backups are necessary to enable you to get back to the last known good state.

The warm standby strategy deploys a functional stack, but at reduced capacity. The DR endpoint can handle requests, but cannot handle production levels of traffic. It may be more, but is always less than the full production deployment for cost savings. If the passive stack is deployed to the recovery Region at full capacity, however, then this strategy is known as "hot standby." Because warm standby

deploys a functional stack to the recovery Region, this makes it easier to test Region readiness using synthetic transactions.

Disaster recovery (DR) strategies:

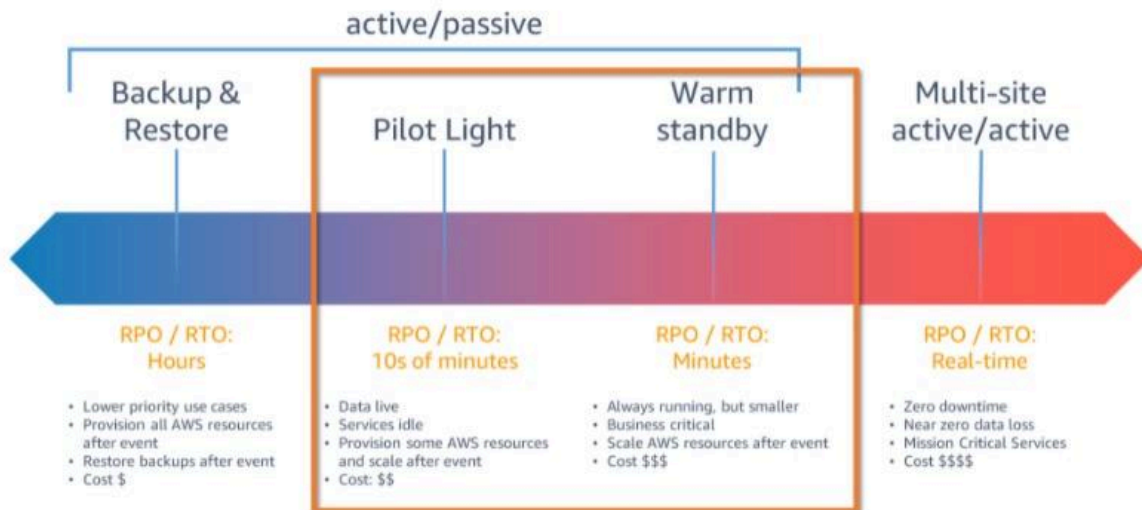


Figure 1. DR strategies

via -

<https://aws.amazon.com/blogs/architecture/disaster-recovery-dr-architecture-on-aws-part-iii-pilot-light-and-warm-standby/>

Incorrect options:

Multi-site active-active strategy - This strategy uses AWS Regions as your active sites, creating a multi-Region active/active architecture. Generally, two Regions are used. Each Region hosts a highly available, multi-Availability Zone (AZ) workload stack. In each Region, data is replicated live between the data stores and also backed up. This protects against disasters that include data deletion or corruption since the data backup can be restored to the last known good state. Each regional stack serves production traffic effectively. But, this strategy is cost involving and should only be used for mission-critical applications.

Pilot Light strategy - Pilot Light, like Warm Standby strategy, replicates data from the primary Region to data resources in the recovery Region, such as Amazon Relational Database Service (Amazon RDS) DB instances or Amazon DynamoDB tables. But, the DR Region in a pilot light strategy (unlike warm standby) cannot serve requests until additional steps are taken. A pilot light in a home furnace does not provide heat

to the home. It provides a quick way to light the furnace burners that then provide heat.

Warm standby can handle traffic at reduced levels immediately. Pilot light requires you to first deploy infrastructure and then scale out resources before the workload can handle requests.

Backup & Restore strategy - Backup and Restore is associated with higher RTO (recovery time objective) and RPO (recovery point objective). This results in longer downtimes and greater loss of data between when the disaster event occurs and recovery. However, backup and restore can still be the right strategy for workloads because it is the easiest and least expensive strategy to implement.

Reference:

<https://aws.amazon.com/blogs/architecture/disaster-recovery-dr-architecture-on-aws-part-iii-pilot-light-and-warm-standby/>

Domain

Cloud Concepts

Question 24Correct

Which of the following is correct regarding the Amazon Relational Database Service (Amazon RDS) service?

Your answer is correct

You can use both read replicas and multi-AZ deployment for disaster recovery

You can use both read replicas and multi-AZ deployment having single standby for improved read performance

You can use read replicas for disaster recovery only and multi-AZ deployment for improved read performance only

You can use read replicas for improved read performance only and multi-AZ deployment for disaster recovery only

Overall explanation

Correct option:

You can use both read replicas and multi-AZ deployment for disaster recovery

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. Read replicas allow you to create read-only copies that are synchronized with your master database. Read replicas are used for improved read performance. You can also place your read replica in a

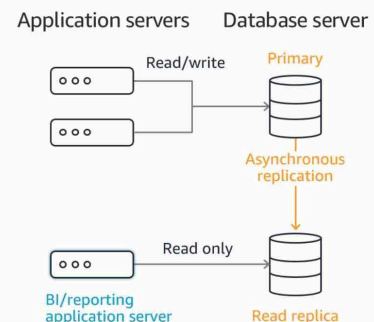
different AWS Region closer to your users for better performance. Using a cross-Region read replica can also help ensure that you get back up and running if you experience a regional availability issue in case of a disaster. Read replicas are an example of horizontal scaling of resources.

Read replica Overview:

Amazon RDS Read Replicas provide enhanced performance and durability for RDS database (DB) instances. They make it easy to elastically scale out beyond the capacity constraints of a single DB instance for read-heavy database workloads. You can create one or more replicas of a given source DB Instance and serve high-volume application read traffic from multiple copies of your data, thereby increasing aggregate read throughput. Read replicas can also be promoted when needed to become standalone DB instances. Read replicas are available in Amazon RDS for [MySQL](#), [MariaDB](#), [PostgreSQL](#), [Oracle](#), and [SQL Server](#) as well as Amazon Aurora.

For the MySQL, MariaDB, PostgreSQL, Oracle, and SQL Server database engines, Amazon RDS creates a second DB instance using a snapshot of the source DB instance. It then uses the engines' native asynchronous replication to update the read replica whenever there is a change to the source DB instance. The read replica operates as a DB instance that allows only read-only connections; applications can connect to a read replica just as they would to any DB instance. Amazon RDS replicates all databases in the source DB instance.

[Amazon Aurora](#) further extends the benefits of read replicas by employing an SSD-backed virtualized storage layer purpose-built for database workloads. Amazon Aurora replicas share the same underlying storage as the source instance, lowering costs and avoiding the need to copy data to the replica nodes. For more information about replication with Amazon Aurora, see the [online documentation](#).



via - <https://aws.amazon.com/rds/features/multi-az/>

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for RDS database (DB) instances, making them a natural fit for production database workloads. When you provision an Amazon RDS Multi-AZ DB Instance, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ). Amazon RDS performs an automatic failover to the standby so that you can resume database operations as soon as the failover is complete. Since the endpoint for your DB Instance remains the same after a failover, your application can resume database operation without the need for manual administrative intervention. Think of multi-AZ deployment as enhancing the availability and reliability of your system. For a disaster event based on disruption or loss of one physical data center, implementing a highly available workload in multiple Availability Zones within a single AWS Region helps mitigate against natural and technical disasters. So, RDS Multi-AZ can be used to address DR requirements.

In addition, Amazon RDS Multi-AZ with two readable standbys maximizes read-performance and scalability.

To understand the Amazon Relational Database Service (Amazon RDS) disaster recovery capabilities in more detail, you can refer to this excellent AWS blog:

<https://aws.amazon.com/blogs/database/implementing-a-disaster-recovery-strategy-with-amazon-rds/>

Incorrect options:

You can use read replicas for improved read performance only and multi-AZ deployment for disaster recovery only

You can use read replicas for disaster recovery only and multi-AZ deployment for improved read performance only

These two options contradict the details provided earlier in the explanation, so these options are incorrect.

You can use both read replicas and multi-AZ deployment having single standby for improved read performance - Amazon RDS Multi-AZ with single standby can only be used to enhance durability and availability. It cannot be used to improve the read performance.

References:

<https://aws.amazon.com/blogs/database/implementing-a-disaster-recovery-strategy-with-amazon-rds/>

<https://aws.amazon.com/rds/features/multi-az/>

Domain

Technology

Question 25Correct

Which of the following describes an Availability Zone (AZ) in the AWS Cloud?

One or more server racks in the same location

One or more server racks in multiple locations

One or more data centers in multiple locations

Your answer is correct

One or more data centers in the same location

Overall explanation

Correct option:

One or more data centers in the same location

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. An AWS Region refers to a physical

location around the world where AWS clusters data centers. AZs give customers the ability to operate production applications and databases that are more highly available, fault-tolerant, and scalable than would be possible from a single data center. All AZs in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZs.

AWS Regions and Availability Zones (AZs) Explained:

Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of multiple, isolated, and physically separate AZ's within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZ's to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

Availability Zones

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. AZ's give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. All AZ's in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZ's. All traffic between AZ's is encrypted. The network performance is sufficient to accomplish synchronous replication between AZ's. AZ's make partitioning applications for high availability easy. If an application is partitioned across AZ's, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZ's are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other.

via - https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Incorrect options:

One or more data centers in multiple locations

One or more server racks in the same location

One or more server racks in multiple locations

These three options contradict the details provided earlier in the explanation, so these options are incorrect.

Reference:

https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Domain

Security and Compliance

Question 26Incorrect

Which of the following is best-suited for load-balancing HTTP and HTTPS traffic?

Your answer is incorrect

Network Load Balancer

AWS Auto Scaling

Correct answer

Application Load Balancer

System Load Balancer

Overall explanation

Correct option:

Application Load Balancer

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions. It can handle the varying load of your application traffic in a single Availability Zone or across multiple Availability Zones. Elastic Load Balancing (ELB) offers three types of load balancers that all feature the high availability, automatic scaling, and robust security necessary to make your applications fault-tolerant.

Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of multiple, isolated, and physically separate AZ's within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZ's to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

Availability Zones

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. AZ's give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. All AZ's in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZ's. All traffic between AZ's is encrypted. The network performance is sufficient to accomplish synchronous replication between AZ's. AZ's make partitioning applications for high availability easy. If an application is partitioned across AZ's, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZ's are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other.

via - <https://aws.amazon.com/elasticloadbalancing/>

Application Load Balancer is used for load balancing of HTTP and HTTPS traffic and provides advanced request routing targeted at the delivery of modern application architectures, including microservices and containers.

Incorrect options:

Network Load Balancer - Network Load Balancer is best suited for load balancing of Transmission Control Protocol (TCP), User Datagram Protocol (UDP) and Transport Layer Security (TLS) traffic where extreme performance is required.

AWS Auto Scaling - AWS Auto Scaling monitors your applications and automatically adjusts the capacity to maintain steady, predictable performance at the lowest possible cost. Using AWS Auto Scaling, it's easy to setup application scaling for multiple resources across multiple services in minutes. The service provides a simple, powerful user interface that lets you build scaling plans for resources including Amazon EC2 instances and Spot Fleets, Amazon ECS tasks, Amazon DynamoDB tables and indexes, and Amazon Aurora Replicas. Auto Scaling cannot be used for load-balancing HTTP and HTTPS traffic.

System Load Balancer - This is a made-up option and has been added as a distractor.

Reference:

<https://aws.amazon.com/elasticloadbalancing/>

Domain

Technology

Question 27Incorrect

AWS Trusted Advisor can provide alerts on which of the following common security misconfigurations? (Select two)?

When you don't tag objects in Amazon S3 buckets

Your selection is incorrect

When you share IAM user credentials with others

Your selection is correct

When you allow public access to Amazon S3 buckets

When you don't enable data encryption on Amazon S3 Glacier

Correct selection

When you don't turn on user activity logging (AWS CloudTrail)

Overall explanation

Correct options:

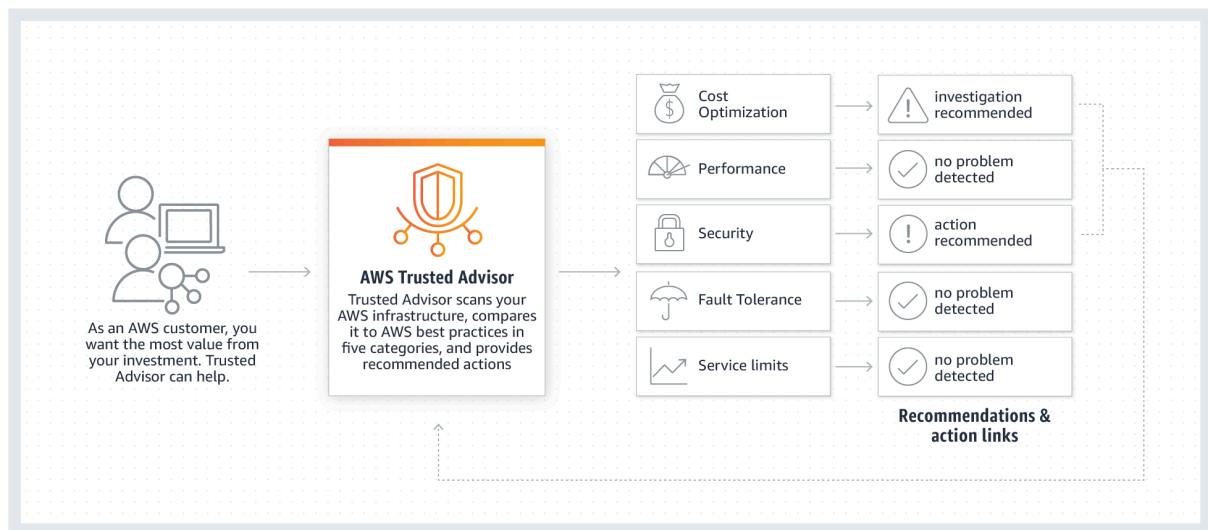
When you allow public access to Amazon S3 buckets

When you don't turn on user activity logging (AWS CloudTrail)

AWS Trusted Advisor is an online tool that provides real-time guidance to help provision your resources following AWS best practices. Whether establishing new workflows, developing applications, or as part of ongoing improvement, recommendations provided by Trusted Advisor regularly help keep your solutions provisioned optimally. AWS Trusted Advisor analyzes your AWS environment and provides best practice recommendations in five categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits.

Trusted Advisor inspects your AWS environment and makes recommendations when opportunities may exist to save money, improve system performance, or close security gaps. It provides alerts on several of the most common security misconfigurations that can occur, including leaving certain ports open that make you vulnerable to hacking and unauthorized access, neglecting to create IAM accounts for your internal users, allowing public access to Amazon S3 buckets, not turning on user activity logging (AWS CloudTrail), or not using MFA on your root AWS Account.

How AWS Trusted Advisor Works:



via - <https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Incorrect options:

When you don't tag objects in Amazon S3 buckets - Tagging objects (or any resource) in Amazon S3 is not mandatory and it's not a security threat.

When you share IAM user credentials with others - It is the customer's responsibility to adhere to the IAM security best practices and never share the IAM user credentials with others. AWS Trusted Advisor cannot send an alert for such use-cases.

When you don't enable data encryption on Amazon S3 Glacier - By default, data is encrypted on S3 Glacier. So, this option has been added as a distractor.

Reference:

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/best-practice-checklist/>

Domain

Security and Compliance

Question 28 Correct

Which of the following are the serverless computing services offered by AWS ?
(Select two)

AWS Elastic Beanstalk

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Lightsail

Your selection is correct

AWS Fargate

Your selection is correct

AWS Lambda

Overall explanation

Correct options:

Serverless is the native architecture of the cloud that enables you to shift more of your operational responsibilities to AWS, increasing your agility and innovation.

Serverless allows you to build and run applications and services without thinking about servers. It eliminates infrastructure management tasks such as server or cluster provisioning, patching, operating system maintenance, and capacity provisioning.

The AWS serverless platform overview:

Compute AWS Lambda lets you run code without provisioning or managing servers. You pay only for the compute time you consume - there is no charge when your code is not running. Lambda@Edge allows you to run Lambda functions at AWS Edge locations in response to Amazon CloudFront events. AWS Fargate is a purpose-built serverless compute engine for containers. Fargate scales and manages the infrastructure required to run your containers.	Storage Amazon Simple Storage Service (Amazon S3) provides developers and IT teams with secure, durable, highly-scalable object storage. Amazon S3 is easy to use, with a simple web service interface to store and retrieve any amount of data from anywhere on the web. Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage. It is built to elastically scale on demand, growing and shrinking automatically as you add and remove files.	Data stores Amazon DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale. Amazon Aurora Serverless is an on-demand, auto-scaling configuration for Amazon Aurora (MySQL-compatible edition), where the database will automatically start up, shut down, and scale capacity up or down based on your application's needs. Amazon RDS Proxy is a highly available database proxy that manages thousands of concurrent connections to relational databases, allowing you to build highly scalable,	API Proxy Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale. It offers a comprehensive platform for API management . API Gateway allows you to process hundreds of thousands of concurrent API calls and handles traffic management, authorization and access control, monitoring, and API version management.
Application integration Amazon SNS is a fully managed pub/sub messaging service that makes it easy to decouple and scale microservices, distributed systems, and serverless applications. Amazon SQS is a fully managed message queuing service that makes it easy to decouple and scale microservices, distributed systems, and serverless applications. AWS AppSync simplifies application development by letting you create a flexible GraphQL API to securely access, manipulate, and combine data from one or more data sources.	Orchestration AWS Step Functions makes it easy to coordinate the components of distributed applications and microservices using visual workflows. Building applications from individual components that each perform a discrete function lets you scale and change applications quickly. Step Functions is a reliable way to coordinate components and step through the functions of your application.	Analytics Amazon Kinesis is a platform for streaming data on AWS, offering powerful services to make it easy to load and analyze streaming data, and also providing the ability for you to build custom streaming data applications for specialized needs. Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL. Athena is serverless, so there is no infrastructure to manage, and you pay only for the queries that you run.	Developer tooling AWS provides tools and services that aid developers in the serverless application development process. AWS and its partner ecosystem offer tools for continuous integration and delivery, testing, deployments, monitoring and diagnostics, SDKs, frameworks, and integrated development environment (IDE) plugins.

via - <https://aws.amazon.com/serverless/>

AWS Lambda

With AWS Lambda, you can run code for virtually any type of application or backend service - all with zero administration. Just upload your code and Lambda takes care of everything required to run and scale your code with high availability. You can set up your code to automatically trigger from other AWS services or call it directly from any web or mobile app.

AWS Lambda lets you run code without provisioning or managing servers. You pay only for the compute time you consume - there is no charge when your code is not running.

AWS Fargate

AWS Fargate is a serverless compute engine for containers that works with both Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS). Fargate makes it easy for you to focus on building your applications. Fargate removes the need to provision and manage servers, lets you specify and pay for resources per application, and improves security through application isolation by design.

AWS Fargate is a purpose-built serverless compute engine for containers. Fargate scales and manages the infrastructure required to run your containers.

Incorrect options:

Amazon Elastic Compute Cloud (Amazon EC2) - Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud with support for per-second billing. It is the easiest way to provision servers on AWS Cloud and access the underlying OS. Amazon EC2 reduces the time required to obtain and boot new server instances to minutes, allowing you to quickly scale capacity, both up and down, as your computing requirements change.

AWS Elastic Beanstalk - AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services. You simply upload your code and Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring. Beanstalk provisions servers so it is not a serverless service.

Amazon Lightsail - Amazon Lightsail is an easy-to-use cloud platform that offers you everything needed to build an application or website, plus a cost-effective, monthly plan. Lightsail offers several preconfigured, one-click-to-launch operating systems, development stacks, and web applications, including Linux, Windows OS, and WordPress.

References:

<https://aws.amazon.com/serverless/>

<https://aws.amazon.com/fargate/>

Domain

Cloud Concepts

Question 29

Reserved Instance (RI) pricing is available for which of the following AWS services? (Select two)

Your selection is correct

Amazon Relational Database Service (Amazon RDS)

Amazon Simple Storage Service (Amazon S3)

AWS Identity and Access Management (AWS IAM)

Amazon CloudFront

Your selection is correct

Amazon Elastic Compute Cloud (Amazon EC2)

Overall explanation

Correct options:

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Relational Database Service (Amazon RDS)

A Reserved Instance (RI) is a reservation that provides a discounted hourly rate in exchange for an upfront fee and term contract. Services such as Amazon Elastic Compute Cloud (Amazon EC2) and Amazon Relational Database Service (Amazon RDS) use this approach to sell reserved capacity for hourly use of Reserved Instances (RI). It is not a virtual machine. It is a commitment to pay in advance for specific Amazon EC2 or Amazon RDS instances.

Incorrect options:

Amazon CloudFront - Amazon CloudFront is a content delivery network (CDN) service. Amazon CloudFront does not offer "Reserved Capacity" pricing.

Amazon Simple Storage Service (Amazon S3) - Amazon S3 infrastructure is managed by AWS. So, Reserved Instance (RI) does not make sense here. But, Amazon S3 offers volume discounts for its storage classes.

AWS Identity and Access Management (AWS IAM) - AWS Identity and Access Management (AWS IAM) enables you to manage access to AWS services and resources securely. Using AWS Identity and Access Management (AWS IAM), you can create and manage IAM users and IAM user groups, and use permissions to allow and deny their access to AWS resources. This is a free service to every AWS customer.

Reference:

<https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/con-bill-blended-rates.html>

Domain

Billing and Pricing

Question 30Incorrect

Which of the following entities should be used for an Amazon Elastic Compute Cloud (Amazon EC2) Instance to access a DynamoDB table?

AWS IAM user access keys

Correct answer

IAM role

Amazon Cognito

Your answer is incorrect

AWS Key Management Service (KMS)

Overall explanation

Correct option:

IAM role

An IAM Role is an IAM identity that you can create in your account that has specific permissions. An IAM role is similar to an IAM user in that it is an AWS identity with permissions policies that determine what the identity can and cannot do in AWS. When you assume a role, it provides you with temporary security credentials for your role session.

Incorrect options:

AWS IAM user access keys - Access keys are long-term credentials for an IAM user or the AWS account root user. You can use access keys to sign programmatic requests to the AWS CLI or AWS API (directly or using the AWS SDK). Access keys consist of two parts: an access key ID and a secret access key. As a user name and password, you must use both the access key ID and secret access key together to authenticate your requests. As a best practice, AWS suggests the use of temporary security credentials (IAM roles) instead of access keys.

Amazon Cognito - Amazon Cognito lets you add user sign-up, sign-in, and access control to your web and mobile apps quickly and easily. Amazon Cognito scales to millions of users and supports sign-in with social identity providers, such as Facebook, Google, and Amazon, and enterprise identity providers via SAML 2.0. Amazon Cognito cannot be used to facilitate an Amazon EC2 Instance to access a DynamoDB table.

AWS Key Management Service (KMS) - AWS Key Management Service (KMS) makes it easy for you to create and manage cryptographic keys and control their use across a wide range of AWS services and in your applications. AWS KMS is a secure and resilient service that uses hardware security modules that have been validated under FIPS 140-2, or are in the process of being validated, to protect your keys. AWS KMS cannot be used to facilitate an Amazon EC2 Instance to access a DynamoDB table.

Reference:

https://docs.amazonaws.cn/en_us/amazondynamodb/latest/developerguide/authentication-and-access-control.html

Domain

Security and Compliance

Question 31 Correct

An organization maintains a separate Virtual Private Cloud (VPC) for each of its business units. Two units need to privately share data. Which is the most optimal way of privately sharing data between the two VPCs?

Your answer is correct

VPC peering connection

VPC Endpoint

AWS Site-to-Site VPN

AWS Direct Connect

Overall explanation

Correct option:

VPC peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your VPCs, with a VPC in another AWS account, or with a VPC in a different AWS Region.

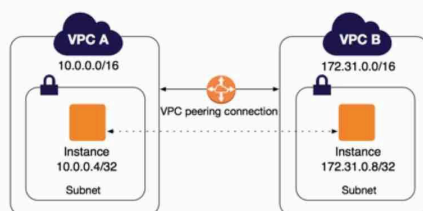
VPC Peering Overview:

What is VPC peering?

[PDF](#)

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that you've defined.

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. The VPCs can be in different regions (also known as an inter-region VPC peering connection).



AWS uses the existing infrastructure of a VPC to create a VPC peering connection; it is neither a gateway nor a VPN connection, and does not rely on a separate piece of physical hardware. There is no single point of failure for communication or a bandwidth bottleneck.

A VPC peering connection helps you to facilitate the transfer of data. For example, if you have more than one AWS account, you can peer the VPCs across those accounts to create a file sharing network. You can also use a VPC peering connection to allow other VPCs to access resources you have in one of your VPCs.

via - <https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

Incorrect options:

AWS Site-to-Site VPN - AWS Site-to-Site VPN creates a secure connection between your data center or branch office and your AWS cloud resources. This connection goes over the public internet. Site to Site VPN cannot be used to interconnect VPCs.

AWS Direct Connect - AWS Direct Connect creates a dedicated private connection from a remote network to your VPC. This is a private connection and does not use

the public internet. Takes at least a month to establish this connection. AWS Direct Connect cannot be used to interconnect VPCs.

VPC Endpoint - A VPC endpoint enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by AWS PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. You cannot connect two VPCs using a VPC endpoint.

Reference:

<https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html>

Domain

Cloud Concepts

Question 32Incorrect

An e-commerce company has migrated its IT infrastructure from the on-premises data center to AWS Cloud. Which of the following costs is the company responsible for?

Costs for hardware infrastructure on AWS Cloud

Costs for powering servers on AWS Cloud

Correct answer

Application software license costs

Your answer is incorrect

AWS Data Center physical security costs

Overall explanation

Correct option:

Application software license costs

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing. With cloud computing, you don't need to make large upfront investments in hardware and spend a lot of time on the heavy lifting of managing that hardware. Therefore, all costs for hardware infrastructure, powering servers and physical security for the Data Center fall under the ambit of AWS.

The customer needs to take care of software licensing costs and human resources costs.

Incorrect options:

AWS Data Center physical security costs

Costs for hardware infrastructure on AWS Cloud

Costs for powering servers on AWS Cloud

As per the details mentioned in the explanation above, these three options are not correct for the given use-case.

Reference:

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/what-is-cloud-computing.html>

Domain

Cloud Concepts

Question 33Incorrect

Which AWS service would you choose for a data processing project that needs a schemaless database?

Amazon Relational Database Service (Amazon RDS)

Correct answer

Amazon DynamoDB

Amazon RedShift

Your answer is incorrect

Amazon Aurora

Overall explanation

Correct option:

Amazon DynamoDB

Amazon DynamoDB is a key-value and document database that delivers single-digit millisecond performance at any scale. It's a fully managed, multi-Region, multi-master, durable database with built-in security, backup and restore, and in-memory caching for internet-scale applications. DynamoDB is schemaless. DynamoDB can manage structured or semistructured data, including JSON documents.

Incorrect options:

Amazon RedShift - Amazon Redshift is a fully-managed petabyte-scale cloud-based data warehouse product designed for large scale data set storage and analysis. Amazon Redshift requires a well-defined schema.

Amazon Aurora - Amazon Aurora is an AWS service for relational databases. Aurora requires a well-defined schema.

Amazon Relational Database Service (Amazon RDS) - Amazon Relational Database Service (Amazon RDS) is an AWS service for relational databases. RDS requires a well-defined schema.

References:

<https://aws.amazon.com/dynamodb/features/>

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/SQLtoNoSQL.WhyDynamoDB.html>

Domain

Technology

Question 34Incorrect

A multi-national organization has a separate virtual private cloud (VPC) for each of its business units on the AWS Cloud. The organization also wants to connect its on-premises data center with the different virtual private clouds (VPC) for better organization-wide collaboration. Which AWS services can be combined to build the MOST efficient solution for this use-case? (Select two)

Your selection is incorrect

VPC peering connection

Correct selection

AWS Transit Gateway

Your selection is correct

AWS Direct Connect

AWS Storage Gateway

Internet Gateway

Overall explanation

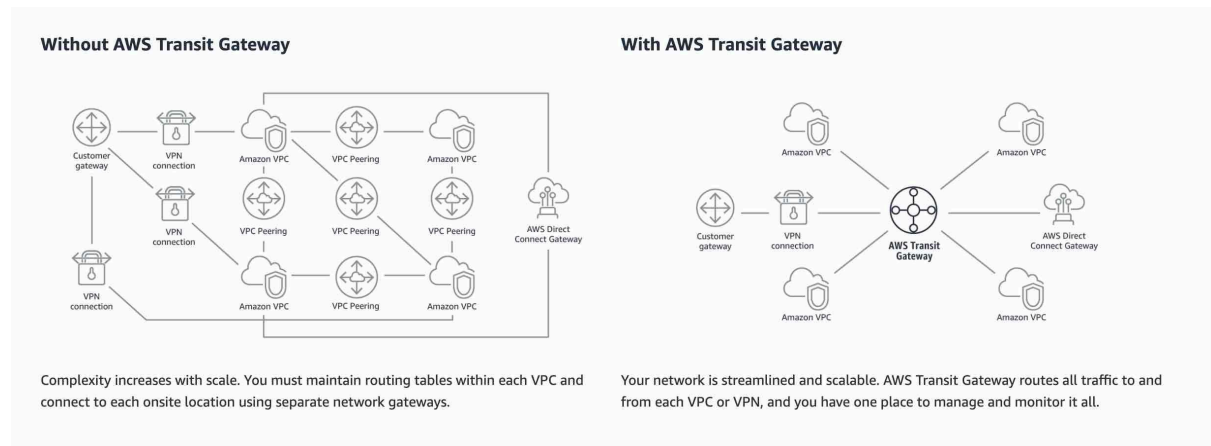
Correct option:

AWS Transit Gateway

AWS Transit Gateway connects virtual private clouds (VPC) and on-premises networks through a central hub. This simplifies your network and puts an end to

complex peering relationships. It acts as a cloud router – each new connection is only made once. As you expand globally, inter-Region peering connects AWS Transit Gateways using the AWS global network. Your data is automatically encrypted and never travels over the public internet.

How AWS Transit Gateway can simplify your network:



via - <https://aws.amazon.com/transit-gateway/>

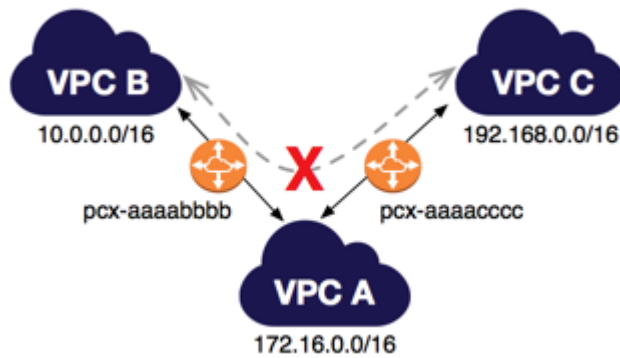
AWS Direct Connect

AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your datacenter, office, or colocation environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based connections.

Incorrect options:

VPC peering connection - A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. VPC peering connection is not transitive, a separate VPC peering connection has to be made between two VPCs that need to talk to each other. With a growing number of virtual private clouds (VPC), this gets difficult to manage.

Transitive VPC peering connection is not allowed:



via -

<https://docs.aws.amazon.com/vpc/latest/peering/invalid-peering-configurations.html>

Internet Gateway - An internet gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet. It, therefore, imposes no availability risks or bandwidth constraints on your network traffic. You cannot use the internet gateway to connect your on-premises data center with multiple VPCs within your AWS network.

AWS Storage Gateway - AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. All data transferred between the gateway and AWS storage is encrypted using SSL (for all three types of gateways - File, Volume and Tape Gateways). You cannot use AWS Storage Gateway to connect your on-premises data center with multiple VPCs within your AWS network.

Reference:

<https://aws.amazon.com/transit-gateway/>

Domain

Cloud Concepts

Question 35Incorrect

Which AWS service can be used to host a static website with the LEAST effort?

AWS Storage Gateway

Amazon Elastic File System (Amazon EFS)

Your answer is incorrect

Amazon S3 Glacier

Correct answer

Amazon Simple Storage Service (Amazon S3)

Overall explanation

Correct option:

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Amazon S3's flat, non-hierarchical structure and various management features are helping customers of all sizes and industries organize their data in ways that are valuable to their businesses and teams. To host a static website on Amazon S3, you configure an Amazon S3 bucket for website hosting and then upload your website content to the bucket. When you configure a bucket as a static website, you must enable website hosting, set permissions, and create and add an index document.

Hosting a static website on Amazon S3:

Hosting a static website on Amazon S3

[PDF](#) | [Kindle](#) | [RSS](#)

You can use Amazon S3 to host a static website. On a *static* website, individual webpages include static content. They might also contain client-side scripts.

By contrast, a *dynamic* website relies on server-side processing, including server-side scripts such as PHP, JSP, or ASP.NET. Amazon S3 does not support server-side scripting, but AWS has other resources for hosting dynamic websites. To learn more about website hosting on AWS, see [Web Hosting](#).

Note

You can use the AWS Amplify Console to host a single page web app. The AWS Amplify Console supports single page apps built with single page app frameworks (for example, React JS, Vue JS, Angular JS, and Nuxt) and static site generators (for example, Gatsby JS, React-static, Jekyll, and Hugo). For more information, see [Getting Started](#) in the *AWS Amplify Console User Guide*.

To configure your bucket for static website hosting, you can use the AWS Management Console without writing any code. You can also create, update, and delete the website configuration *programmatically* by using the AWS SDKs. The SDKs provide wrapper classes around the Amazon S3 REST API. If your application requires it, you can send REST API requests directly from your application.

To host a static website on Amazon S3, you configure an Amazon S3 bucket for website hosting and then upload your website content to the bucket. When you configure a bucket as a static website, you must [enable website hosting](#), [set permissions](#), and [create and add an index document](#). Depending on your website requirements, you can also configure [redirects](#), [web traffic logging](#), and a [custom error document](#).

After you configure your bucket as a static website, you can access the bucket through the AWS Region-specific Amazon S3 website endpoints for your bucket. Website endpoints are different from the endpoints where you send REST API requests. For more information, see [Website endpoints](#).

via - <https://docs.aws.amazon.com/AmazonS3/latest/dev/WebsiteHosting.html>

Incorrect options:

AWS Storage Gateway - AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. Customers use Storage Gateway to simplify storage management and reduce costs for key hybrid cloud storage use cases. It helps on-premises applications to access data on AWS Cloud. It cannot be used to host a website.

Amazon Elastic File System (Amazon EFS) - Amazon Elastic File System (Amazon EFS) provides a simple, scalable, fully managed elastic NFS file system for use with

AWS Cloud services and on-premises resources. EFS storage option cannot directly be used to host a website, EFS needs to be mounted on Amazon EC2 to work as a static website.

Amazon S3 Glacier - Amazon S3 Glacier is a secure, durable, and extremely low-cost Amazon S3 cloud storage class for data archiving and long-term backup. As you see, this cannot be used for hosting a website.

Reference:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/WebsiteHosting.html>

Domain

Technology

Question 36Incorrect

Which of the following AWS Support plans provide programmatic access to AWS Support Center features to create, manage and close your support cases? (Select two)

Your selection is correct

AWS Enterprise Support

AWS Corporate Support

AWS Basic Support

Your selection is incorrect

AWS Developer Support

Correct selection

AWS Business Support

Overall explanation

Correct options:

AWS offers four different support plans to cater to each of its customers - AWS Developer Support, AWS Business Support, AWS Enterprise On-Ramp Support and AWS Enterprise Support plans. A basic support plan is included for all AWS customers.

AWS Enterprise Support

You should use the AWS Enterprise Support plan to provide customers with concierge-like service where the main focus is helping the customer achieve their outcomes and find success in the cloud. With AWS Enterprise Support plan, you get 24x7 technical support from high-quality engineers, tools and technology to

automatically manage the health of your environment, consultative architectural guidance delivered in the context of your applications and use-cases, and a designated Technical Account Manager (TAM) to coordinate access to proactive/preventative programs and AWS subject matter experts. AWS Enterprise Support plan provides 24x7 phone, email and chat access to technical support however it's costlier than the AWS Business Support plan. You get programmatic access (API Access) to AWS Support Center features to create, manage, and close your support cases, and operationally manage your Trusted Advisor check requests and status. You should note that the AWS Enterprise On-Ramp Support plan also gets programmatic access (API Access) to AWS Support Center.

AWS Business Support

You should use the AWS Business Support plan if you have production workloads on AWS and want 24x7 phone, email and chat access to technical support and architectural guidance in the context of your specific use-cases. You get programmatic access (API Access) to AWS Support Center features to create, manage, and close your support cases, and operationally manage your Trusted Advisor check requests and status.

Exam Alert:

AWS Business Support, AWS Enterprise On-Ramp, and AWS Enterprise Support plans support full set of checks for AWS Trusted Advisor best practices and also provides support for programmatic case management.

Please review the differences between the AWS Developer Support, AWS Business Support, AWS Enterprise On-Ramp Support and AWS Enterprise Support plans as you can expect at least a couple of questions on the exam:

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
AWS Trusted Advisor Best Practice Checks	Service Quota and basic Security checks	Full set of checks	Full set of checks	Full set of checks
AWS Trusted Advisor Priority				Prioritized recommendations curated by your AWS account team
Enhanced Technical Support	Business hours** web access to Cloud Support Associates Unlimited cases with 1 primary contact Prioritized responses on AWS re:Post	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack
Case Severity / Response Times*	General guidance: < 24 hours** System impaired: < 12 hours**	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business-critical system down: < 30 minutes	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business/Mission-critical system down: < 15 minutes
Architectural Guidance	General	Contextual to your use-cases	Consultative review and guidance based on your applications (one-per-year)	Consultative review and guidance based on your applications
Programmatic Case Management		AWS Support API	AWS Support API	AWS Support API

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Third-Party Software Support		Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting
Proactive Programs and Self Service	Access to Support Automation Workflows with prefixes AWSsupport	Access to Infrastructure Event Management for additional fee Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Infrastructure Event Management (one-per-year) Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Access to Infrastructure Event Management Access to proactive reviews, workshops, and deep dives Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport
AWS Incident Detection and Response				Access to AWS Incident Detection and Response for an additional fee. AWS Incident Detection and Response is an add-on to Enterprise Support that offers 24x7 proactive monitoring and incident management for selected workloads. AWS Incident Detection and Response leverages the proven operational, enhanced monitoring, and incident management capabilities used internally by AWS teams and externally by AWS Managed Services (AMS).
AWS Managed Services		Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud operations skills and capacity. It includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team. AWS Incident Detection and Response is available at no additional charge in eligible regions for AWS Managed Services direct customers with AWS Enterprise Support.

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Technical Account Management			A pool of Technical Account Managers to provide proactive guidance, and coordinate access to programs and AWS experts	Designated Technical Account Manager (TAM) to proactively monitor your environment and assist with optimization and coordinate access to programs and AWS experts
Training				Access to online self-paced labs
Account Assistance			Concierge Support Team	Concierge Support Team
Pricing	Greater of \$29 / month*** - or - 3% of monthly AWS usage See pricing detail and example.	Greater of \$100 / month*** - or - 10% of monthly AWS usage for the first \$0–\$10K 7% of monthly AWS usage from \$10K–\$80K 5% of monthly AWS usage from \$80K–\$250K 3% of monthly AWS usage over \$250K See pricing detail and example.	Greater of \$5,500 - or - 10% of monthly AWS usage See pricing detail and example.	Greater of \$15,000 - or - 10% of monthly AWS usage for the first \$0–\$150K 7% of monthly AWS usage from \$150K–\$500K 5% of monthly AWS usage from \$500K–\$1M 3% of monthly AWS usage over \$1M See pricing detail and example.
*Additional services for additional fee		*Access to AWS Managed Services (AMS) for an additional fee	*Access to AWS Managed Services (AMS) for an additional fee	* Access to AWS Incident Detection and Response for an additional fee. *Access to AWS Managed Services (AMS) for an additional fee

via - <https://aws.amazon.com/premiumsupport/plans/>

Incorrect options:

AWS Basic Support - AWS Basic Support plan only provides access to the following:

Customer Service & Communities - 24x7 access to customer service, documentation, whitepapers, and support forums. AWS Trusted Advisor - Access to the core Trusted Advisor checks and guidance to provision your resources following best practices to increase performance and improve security. AWS Health Dashboard – Your account health : A personalized view of the health of AWS services, and alerts when your resources are impacted.

AWS Developer Support - You should use the AWS Developer Support plan if you are testing or doing early development on AWS and want the ability to get email based technical support during business hours as well as general architectural guidance as you build and test.

Both these plans do not support programmatic access (API Access) to AWS Support Center.

AWS Corporate Support - This is a made-up option and has been added as a distractor.

Reference:

<https://aws.amazon.com/premiumsupport/plans/>

Domain

Billing and Pricing

Question 37Incorrect

Which of the following statements are CORRECT regarding AWS Global Accelerator?

(Select two)

AWS Global Accelerator uses the AWS global network and its edge locations. But the edge locations used by Global Accelerator are different from Amazon CloudFront edge locations

AWS Global Accelerator cannot be configured with an Elastic Load Balancer (ELB)

Correct selection

AWS Global Accelerator is a good fit for non-HTTP use cases

Your selection is correct

AWS Global Accelerator provides static IP addresses that act as a fixed entry point to your applications

Your selection is incorrect

AWS Global Accelerator can be used to host static websites

Overall explanation

Correct options:

AWS Global Accelerator is a networking service that helps you improve the availability and performance of the applications that you offer to your global users. Global Accelerator improves performance for a wide range of applications over TCP or UDP by proxying packets at the edge to applications running in one or more AWS Regions.

How AWS Global Accelerator Works:



via - <https://aws.amazon.com/global-accelerator/>

AWS Global Accelerator is a good fit for non-HTTP use cases

AWS Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover.

AWS Global Accelerator provides static IP addresses that act as a fixed entry point to your applications

It provides static IP addresses that provide a fixed entry point to your applications and eliminate the complexity of managing specific IP addresses for different AWS Regions and Availability Zones (AZs).

Incorrect options:

AWS Global Accelerator uses the AWS global network and its edge locations. But the edge locations used by Global Accelerator are different from Amazon

CloudFront edge locations - AWS Global Accelerator and Amazon CloudFront use the same edge locations.

AWS Global Accelerator cannot be configured with an Elastic Load Balancer (ELB) -

A regional ELB load balancer is an ideal target for AWS Global Accelerator. AWS Global Accelerator complements ELB by extending these capabilities beyond a single AWS Region, allowing you to provide a global interface for your applications in any number of Regions.

AWS Global Accelerator can be used to host static websites - Amazon S3 can host static websites. So this option is incorrect.

Reference:

<https://aws.amazon.com/global-accelerator/>

Domain

Technology

Question 38Incorrect

Which of the following are recommended security best practices for the AWS account root user? (Select two)

Correct selection

Set up an IAM user with administrator permissions and do not use AWS account root user for administrative tasks

Share AWS account root user access keys with other administrators

Disable multi-factor authentication (MFA) for the AWS account root user as it can lock the entire AWS account if the multi-factor authentication (MFA) device is lost

Your selection is incorrect

Keep your AWS account root user access keys in an encrypted file on Amazon S3

Your selection is correct

Enable multi-factor authentication (MFA) for the AWS account root user

Overall explanation

Correct options:

Enable multi-factor authentication (MFA) for the AWS account root user

Set up an IAM user with administrator permissions and do not use AWS account root user for administrative tasks

When you create an AWS account, you create an AWS account root user identity, which you use to sign in to AWS. You can sign in to the AWS Management Console using this root user identity – that is, the email address and password that you provided when creating the account. This combination of your email address and password is also called your root user credentials.

Some of the AWS account root user security best practices are as follows:

Do not use the AWS account root user for any task where it's not required. Instead, create a new IAM user for each person that requires administrator access. Then make those users administrators by placing the users into an "Administrators" group to which you attach the AdministratorAccess managed policy.

If you don't already have an access key for your AWS account root user, don't create one unless you need to. If you do have an access key for your AWS account root user, delete it.

Never share your AWS account root user password or access keys with anyone. Use a strong password to help protect account-level access to the AWS Management Console.

Enable AWS multi-factor authentication (MFA) on your AWS account root user account.

Lock Away Your AWS Account Root User Access Keys

You use an access key (an access key ID and secret access key) to make programmatic requests to AWS. However, do not use your AWS account root user access key. The access key for your AWS account root user gives full access to all your resources for all AWS services, including your billing information. You cannot reduce the permissions associated with your AWS account root user access key.

Therefore, protect your root user access key like you would your credit card numbers or any other sensitive secret. Here are some ways to do that:

- If you don't already have an access key for your AWS account root user, don't create one unless you absolutely need to. Instead, use your account email address and password to sign in to the AWS Management Console and [create an IAM user for yourself](#) that has administrative permissions.
- If you do have an access key for your AWS account root user, delete it. If you must keep it, rotate (change) the access key regularly. To delete or rotate your root user access keys, go to the [My Security Credentials page](#) in the AWS Management Console and sign in with your account's email address and password. You can manage your access keys in the **Access keys** section. For more information about rotating access keys, see [Rotating Access Keys](#).
- **Never share your AWS account root user password or access keys with anyone.** The remaining sections of this document discuss various ways to avoid having to share your AWS account root user credentials with other users. They also explain how to avoid having to embed them in an application.
- Use a strong password to help protect account-level access to the AWS Management Console. For information about managing your AWS account root user password, see [Changing the AWS Account Root User Password](#).
- **Enable AWS multi-factor authentication (MFA) on your AWS account root user account.** For more information, see [Using Multi-Factor Authentication \(MFA\) in AWS](#).

via -

<https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html#lock-away-credentials>

Incorrect options:

Disable multi-factor authentication (MFA) for the AWS account root user as it can lock the entire AWS account if the multi-factor authentication (MFA) device is lost - AWS recommends that you enable AWS multi-factor authentication (MFA) on your AWS account root user account.

Keep your AWS account root user access keys in an encrypted file on Amazon S3 -

AWS recommends that if you do have an access key for your AWS account root user, delete it.

Share AWS account root user access keys with other administrators - The access key for your AWS account root user gives full access to all your resources for all AWS services, including your billing information. You cannot reduce the permissions associated with your AWS account root user access key. You should never share these access keys with any other users, not even the administrators.

References:

<https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html#lock-away-credentials>

https://docs.aws.amazon.com/IAM/latest/UserGuide/getting-started_create-admin-group.html

Domain

Security and Compliance

Question 39Correct

Which pillar of AWS Well-Architected Framework is responsible for making sure that you select the right resource types and sizes based on your workload requirements?

Cost Optimization

Your answer is correct

Performance Efficiency

Operational Excellence

Reliability

Overall explanation

Correct option:

The AWS Well-Architected Framework helps you understand the pros and cons of decisions you make while building systems on AWS. By using the AWS Well-Architected Framework you will learn architectural best practices for designing and operating reliable, secure, efficient, and cost-effective systems in the cloud. It provides a way for you to consistently measure your architectures against best practices and identify areas for improvement.

The AWS Well-Architected Framework is based on six pillars – Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization and Sustainability.

Overview of the six pillars of the AWS Well-Architected Framework:

AWS Well-Architected and the Six Pillars

Framework Overview

The AWS Well-Architected Framework describes key concepts, design principles, and architectural best practices for designing and running workloads in the cloud. By answering a few foundational questions, learn how well your architecture aligns with cloud best practices and gain guidance for making improvements.

[HTML](#) | [Kindle](#) | [Labs](#)



Operational Excellence Pillar The operational excellence pillar focuses on running and monitoring systems, and continually improving processes and procedures. Key topics include automating changes, responding to events, and defining standards to manage daily operations. HTML Kindle Labs	Security Pillar The security pillar focuses on protecting information and systems. Key topics include confidentiality and integrity of data, managing user permissions, and establishing controls to detect security events. HTML Kindle Labs	Reliability Pillar The reliability pillar focuses on workloads performing their intended functions and how to recover quickly from failure to meet demands. Key topics include distributed system design, recovery planning, and adapting to changing requirements. HTML Kindle Labs
Performance Efficiency Pillar The performance efficiency pillar focuses on structured and streamlined allocation of IT and computing resources. Key topics include selecting resource types and sizes optimized for workload requirements, monitoring performance, and maintaining efficiency as business needs evolve.	Cost Optimization Pillar The cost optimization pillar focuses on avoiding unnecessary costs. Key topics include understanding spending over time and controlling fund allocation, selecting resources of the right type and quantity, and scaling to meet business needs without overspending.	Sustainability Pillar The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads. Key topics include a shared responsibility model for sustainability, understanding impact, and maximizing utilization to minimize required resources and reduce downstream impacts.

via - <https://aws.amazon.com/architecture/well-architected/>

Performance Efficiency - The performance efficiency pillar focuses on using IT and computing resources efficiently. Key topics include selecting the right resource types and sizes based on workload requirements, monitoring performance, and making informed decisions to maintain efficiency as business needs evolve.

Incorrect options:

Cost Optimization - Cost Optimization focuses on avoiding un-needed costs. Key topics include understanding and controlling where the money is being spent, selecting the most appropriate and right number of resource types, analyzing spend over time, and scaling to meet business needs without overspending.

Reliability - This refers to the ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues.

Operational Excellence - The Operational Excellence pillar includes the ability to run and monitor systems to deliver business value and to continually improve supporting processes and procedures. In the cloud, you can apply the same engineering discipline that you use for application code to your entire environment. You can define your entire workload (applications, infrastructure) as code and update it with code. You can implement your operations procedures as code and automate their execution by triggering them in response to events.

Reference:

<https://aws.amazon.com/architecture/well-architected/>

Domain

Cloud Concepts

Question 40Incorrect

A firm wants to maintain the same data on Amazon Simple Storage Service (Amazon S3) between its production account and multiple test accounts. Which technique should you choose to copy data into multiple test accounts while retaining object metadata?

Amazon S3 Transfer Acceleration (Amazon S3TA)

Amazon S3 Storage Classes

Your answer is incorrect

Amazon S3 Bucket Policy

Correct answer

Amazon S3 Replication

Overall explanation

Correct option:

Amazon S3 Replication

Replication enables automatic, asynchronous copying of objects across Amazon S3 buckets. Buckets that are configured for object replication can be owned by the same AWS account or by different accounts. You can copy objects between different AWS Regions or within the same Region. You can use replication to make copies of your objects that retain all metadata, such as the original object creation time and version IDs. This capability is important if you need to ensure that your replica is identical to the source object.

Exam Alert:

Amazon S3 supports two types of replication: S3 cross-region replication (S3 CRR) vs S3 same-region replication (S3 SRR). Please review the differences between S3 SRR and S3 CRR:

When to Use CRR

Cross-Region replication can help you do the following:

- **Meet compliance requirements** — Although Amazon S3 stores your data across multiple geographically distant Availability Zones by default, compliance requirements might dictate that you store data at even greater distances. Cross-Region replication allows you to replicate data between distant AWS Regions to satisfy these requirements.
- **Minimize latency** — If your customers are in two geographic locations, you can minimize latency in accessing objects by maintaining object copies in AWS Regions that are geographically closer to your users.
- **Increase operational efficiency** — If you have compute clusters in two different AWS Regions that analyze the same set of objects, you might choose to maintain object copies in those Regions.

When to Use SRR

Same-Region replication can help you do the following:

- **Aggregate logs into a single bucket** — If you store logs in multiple buckets or across multiple accounts, you can easily replicate logs into a single, in-Region bucket. This allows for simpler processing of logs in a single location.
- **Configure live replication between production and test accounts** — If you or your customers have production and test accounts that use the same data, you can replicate objects between those multiple accounts, while maintaining object metadata, by implementing SRR rules.
- **Abide by data sovereignty laws** — You might be required to store multiple copies of your data in separate AWS accounts within a certain Region. Same-Region replication can help you automatically replicate critical data when compliance regulations don't allow the data to leave your country.

via - <https://docs.aws.amazon.com/AmazonS3/latest/dev/replication.html>

Incorrect options:

Amazon S3 Bucket Policy - A bucket policy is a resource-based AWS Identity and Access Management (IAM) policy. You add a bucket policy to a bucket to grant other AWS accounts or IAM users access permissions for the bucket and the objects in it. Object permissions apply only to the objects that the bucket owner creates. You cannot replicate data using a bucket policy.

Amazon S3 Transfer Acceleration (Amazon S3TA) - Amazon S3 Transfer Acceleration (Amazon S3TA) enables fast, easy, and secure transfers of files over long distances between your client and an S3 bucket. Transfer Acceleration takes

advantage of Amazon CloudFront's globally distributed edge locations. This facility speeds up access between end-user and S3, this is not for replicating data.

Amazon S3 Storage Classes - Amazon S3 offers a range of storage classes designed for different use cases. Each storage class has a defined set of rules to store, encrypt data at a certain price. Based on the use case, customers can choose the storage class that best suits their business requirements.

These include S3 Standard for general-purpose storage of frequently accessed data; S3 Intelligent-Tiering for data with unknown or changing access patterns; S3 Standard-Infrequent Access (S3 Standard-IA) and S3 One Zone-Infrequent Access (S3 One Zone-IA) for long-lived, but less frequently accessed data; and Amazon S3 Glacier (S3 Glacier) and Amazon S3 Glacier Deep Archive (S3 Glacier Deep Archive) for long-term archive and digital preservation. You cannot replicate data using storage classes.

Reference:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/replication.html>

Domain

Technology

Question 41Incorrect

According to the AWS Shared Responsibility Model, which of the following are responsibilities of the customer for AWS Identity and Access Management (AWS IAM)? (Select two)

Your selection is correct

Analyze user access patterns and review AWS Identity and Access Management (AWS IAM) permissions

Your selection is incorrect

Manage global network security infrastructure

Compliance validation for the underlying software infrastructure

Correct selection

Enable multi-factor authentication (MFA) on all accounts

Configuration and vulnerability analysis for the underlying software infrastructure

Overall explanation

Correct options:

Security and Compliance is a shared responsibility between AWS and the customer. This shared model can help relieve the customer's operational burden as AWS

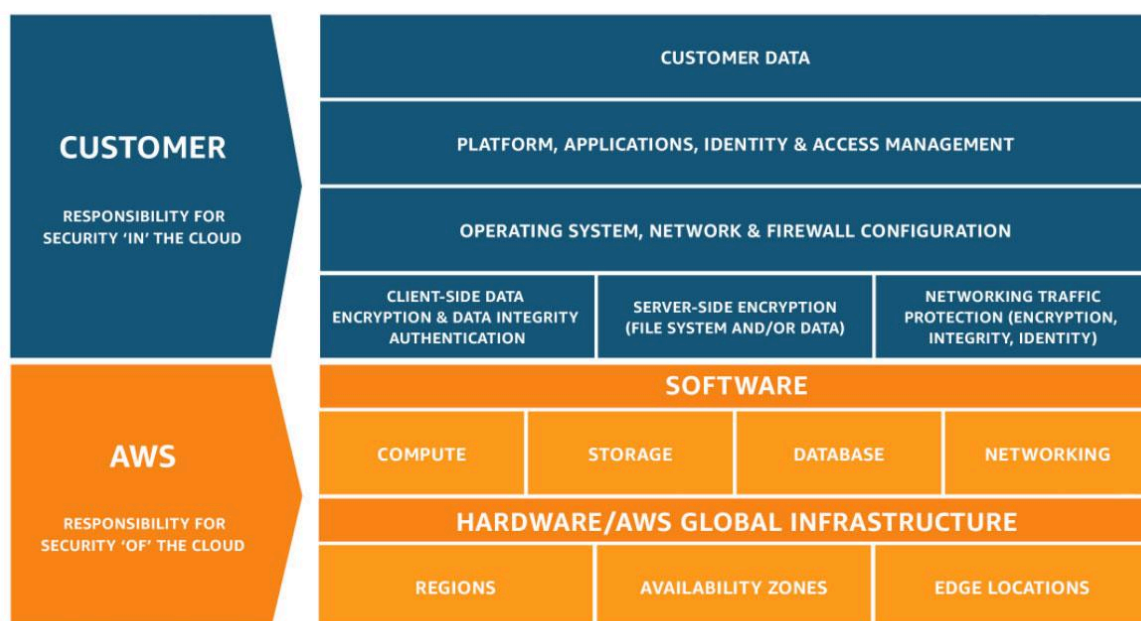
operates, manages and controls the components from the host operating system and virtualization layer down to the physical security of the facilities in which the service operates.

Enable multi-factor authentication (MFA) on all accounts

Analyze user access patterns and review AWS Identity and Access Management (AWS IAM) permissions

Under the AWS Shared Responsibility Model, customers are responsible for enabling multi-factor authentication (MFA) on all accounts, analyzing access patterns and reviewing permissions for AWS Identity and Access Management (AWS IAM) entities.

AWS Shared Responsibility Model Overview:



via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect options:

Manage global network security infrastructure

Configuration and vulnerability analysis for the underlying software infrastructure

Compliance validation for the underlying software infrastructure

According to the AWS Shared Responsibility Model, AWS is responsible for "Security of the Cloud". This includes protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services. Therefore these three options fall under the responsibility of AWS.

Reference:

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Domain

Security and Compliance

Question 42Incorrect

Which of the following AWS entities lists all users in your account and the status of their various account aspects such as passwords, access keys, and MFA devices?

Amazon Inspector

Correct answer

Credentials Report

Your answer is incorrect

AWS Trusted Advisor

AWS Cost & Usage Report (AWS CUR)

Overall explanation

Correct option:

Credentials Report

You can generate and download a credentials report that lists all users in your account and the status of their various credentials, including passwords, access keys, and MFA devices. You can use the credentials report to assist in your auditing and compliance efforts. You can use the report to audit the effects of credential lifecycle requirements, such as password and access key rotation. You can provide the report to an external auditor, or grant permissions to an auditor so that he or she can download the report directly.

Incorrect options:

AWS Trusted Advisor - AWS Trusted Advisor is an online tool that provides real-time guidance to help provision your resources following AWS best practices. Whether establishing new workflows, developing applications, or as part of ongoing improvement, recommendations provided by Trusted Advisor regularly help keep

your solutions provisioned optimally. AWS Trusted Advisor analyzes your AWS environment and provides best practice recommendations in five categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits.

AWS Cost & Usage Report (AWS CUR) - The AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can use the AWS Cost & Usage Report (AWS CUR) to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself. The AWS Cost & Usage Report (AWS CUR) cannot be used to identify under-utilized Amazon EC2 instances.

Amazon Inspector - Amazon Inspector is an automated, security assessment service that helps you check for unintended network accessibility of your Amazon EC2 instances and for vulnerabilities on those EC2 instances. Amazon Inspector assessments are offered to you as pre-defined rules packages mapped to common security best practices and vulnerability definitions.

Reference:

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_getting-report.html

Domain

Security and Compliance

Question 43

Incorrect

Which of the following Amazon Simple Storage Service (Amazon S3) storage classes do not charge any data retrieval fee? (Select two)

Correct selection

Amazon S3 Intelligent-Tiering

Your selection is correct

Amazon S3 Standard

Your selection is incorrect

Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

Amazon S3 Glacier Flexible Retrieval

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

Overall explanation

Correct options:

Amazon S3 Standard

Amazon S3 Standard offers high durability, availability, and performance object storage for frequently accessed data. Amazon S3 Standard offers low latency and high throughput performance. It is designed for durability of 99.999999999% of objects across multiple Availability Zones (AZ). Amazon S3 Standard does not charge any data retrieval fee.

Amazon S3 Intelligent-Tiering

The Amazon S3 Intelligent-Tiering storage class is designed to optimize costs by automatically moving data to the most cost-effective access tier, without performance impact or operational overhead. It works by storing objects in two access tiers: one tier that is optimized for frequent access and another lower-cost tier that is optimized for infrequent access. Amazon S3 Intelligent-Tiering does not charge any data retrieval fee.

Please review this illustration for the Amazon S3 Storage Classes retrieval fee. You don't need to memorize the actual numbers, just remember that Amazon S3 Standard and Amazon S3 Intelligent-Tiering do not charge any retrieval fee:

Performance across the S3 Storage Classes							
	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier Instant Retrieval	S3 Glacier Flexible Retrieval	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128 KB	128 KB	128 KB	40 KB	40 KB
Minimum storage duration charge	N/A	N/A	30 days	30 days	90 days	90 days	180 days
Retrieval charge	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	milliseconds	milliseconds	milliseconds	milliseconds	minutes or hours	hours
Storage type	Object	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes	Yes

via - <https://aws.amazon.com/s3/storage-classes/>

Incorrect options:

Amazon S3 Glacier Flexible Retrieval - Amazon S3 Glacier Flexible Retrieval is a secure, durable, and extremely low-cost Amazon S3 cloud storage class for data archiving and long-term backup. It is designed to deliver 99.999999999% durability,

and provide comprehensive security and compliance capabilities that can help meet even the most stringent regulatory requirements. Amazon S3 Glacier Flexible Retrieval has a data retrieval fee.

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) - Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is for data that is accessed less frequently, but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZ), Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) stores data in a single Availability Zone (AZ). It is not suitable for data archival. Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) has a data retrieval fee.

Amazon S3 Standard-Infrequent Access (S3 Standard-IA) - Amazon S3 Standard-Infrequent Access (S3 Standard-IA) is for data that is accessed less frequently, but requires rapid access when needed. Amazon S3 Standard-Infrequent Access (S3 Standard-IA) offers high durability, high throughput, and low latency of S3 Standard, with a low per GB storage price and per GB retrieval fee. This combination of low cost and high performance makes Amazon S3 Standard-Infrequent Access (S3 Standard-IA) ideal for long-term storage, backups, and as a data store for disaster recovery files. Amazon S3 Standard-Infrequent Access (S3 Standard-IA) has a data retrieval fee.

Reference:

<https://aws.amazon.com/s3/storage-classes/>

Domain

Technology

Question 44Correct

Which AWS service will help you deploy application code automatically to an Amazon Elastic Compute Cloud (Amazon EC2) instance?

AWS CloudFormation

AWS CodeBuild

AWS Elastic Beanstalk

Your answer is correct

AWS CodeDeploy

Overall explanation

Correct option:

AWS CodeDeploy

AWS CodeDeploy is a service that automates application deployments to a variety of compute services including Amazon EC2, AWS Fargate, AWS Lambda, and on-premises instances. CodeDeploy fully automates your application deployments eliminating the need for manual operations. CodeDeploy protects your application from downtime during deployments through rolling updates and deployment health tracking.

Incorrect options:

AWS Elastic Beanstalk - AWS Elastic Beanstalk is the fastest and simplest way to get web applications up and running on AWS. Developers simply upload their application code and the service automatically handles all the details such as resource provisioning, load balancing, auto-scaling, and monitoring. Elastic Beanstalk is an end-to-end application platform, unlike CodeDeploy, which is targeted at code deployment automation for any environment (Development, Testing, Production). It cannot be used to automatically deploy code to an Amazon EC2 instance.

AWS CloudFormation - AWS CloudFormation provides a common language for you to model and provision AWS and third-party application resources in your cloud environment. AWS CloudFormation allows you to use programming languages or a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all regions and accounts. It cannot be used to automatically deploy code to an Amazon EC2 instance.

AWS CodeBuild - AWS CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages that are ready to deploy. With CodeBuild, you don't need to provision, manage, and scale your own build servers. CodeBuild scales continuously and processes multiple builds concurrently, so your builds are not left waiting in a queue. It cannot be used to automatically deploy code to an Amazon EC2 instance.

Reference:

<https://aws.amazon.com/codedeploy/>

Domain

Technology

Question 45Incorrect

How is Amazon EC2 different from traditional hosting systems? (Select two)

With Amazon EC2, users risk overbuying resources

Your selection is incorrect

Amazon EC2 provides a pre-configured instance for a fixed monthly cost

Amazon EC2 caters more towards groups of users with similar system requirements so that the server resources are shared across multiple users and the cost is reduced

Correct selection

Amazon EC2 can scale with changing computing requirements

Your selection is correct

With Amazon EC2, developers can launch and terminate the instances anytime they need to

Overall explanation

Correct options:

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud with support for per-second billing. It is the easiest way to provision servers on AWS Cloud and access the underlying OS.

Amazon EC2 differs fundamentally with the traditional on-premises hosting systems in the flexibility, control and significant cost savings it offers developers, allowing them to treat Amazon EC2 instance as their own customized server backed by the robust infrastructure of AWS Cloud.

Amazon EC2 can scale with changing computing requirements

When computing requirements unexpectedly change, Amazon EC2 can be scaled to match the requirements. Developers can control how many EC2 instances are in use at any given point in time.

With Amazon EC2, developers can launch and terminate the instances anytime they need to

Using Amazon EC2, developers can choose not only to launch, terminate, start or shut down instances at any time, but they can also completely customize the configuration of their instances to suit their needs.

Incorrect options:

Amazon EC2 provides a pre-configured instance for a fixed monthly cost - This is an incorrect option. EC2 developers enjoy the benefit of paying only for their actual resource consumption with no monthly or upfront costs. Developers can customize their EC2 instances for their application stack.

With Amazon EC2, users risk overbuying resources - This is an incorrect statement. Users risk overbuying in traditional hosting services where users pay a fixed, up-front fee irrespective of their actual computing power used. With EC2, users pay only for the actual resources consumed.

Amazon EC2 caters more towards groups of users with similar system requirements so that the server resources are shared across multiple users and the cost is reduced - This is an incorrect statement. Resources are not shared between users in EC2, which is why the users have the flexibility to start or shutdown the instances as per their requirement. This is not possible for the traditional hosting systems where the resources are shared across users.

Reference:

<https://aws.amazon.com/ec2/faqs/>

Domain

Technology

Question 46Incorrect

Which of the following AWS services offers Lifecycle configuration for cost-optimal storage?

Your answer is incorrect

Amazon Elastic Block Store (Amazon EBS)

Amazon EC2 Instance Store

AWS Storage Gateway

Correct answer

Amazon Simple Storage Service (Amazon S3)

Overall explanation

Correct option:

Amazon Simple Storage Service (Amazon S3)

You can manage your objects on Amazon S3 so that they are stored cost-effectively throughout their lifecycle by configuring their Amazon S3 Lifecycle. An Amazon S3

Lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects.

There are two types of actions:

Transition actions — Define when objects transition to another storage class. For example, you might choose to transition objects to the Amazon S3 Standard-IA storage class 30 days after you created them, or archive objects to the Amazon S3 Glacier storage class one year after creating them.

Expiration actions — Define when objects expire. Amazon S3 deletes expired objects on your behalf.

Incorrect options:

Amazon EC2 Instance Store - An Instance Store provides temporary block-level storage for your EC2 instance. This storage is located on disks that are physically attached to the host computer. Instance store is ideal for the temporary storage of information that changes frequently, such as buffers, caches, scratch data, and other temporary content, or for data that is replicated across a fleet of instances, such as a load-balanced pool of web servers. Instance storage is temporary, data is lost if instance experiences failure or is terminated. Instance Store does not offer lifecycle configuration.

Amazon Elastic Block Store (Amazon EBS) - Amazon Elastic Block Store (Amazon EBS) is an easy to use, high-performance block storage service designed for use with Amazon Elastic Compute Cloud (EC2) for both throughput and transaction-intensive workloads at any scale. A broad range of workloads, such as relational and non-relational databases, enterprise applications, containerized applications, big data analytics engines, file systems, and media workflows are widely deployed on Amazon EBS. It does not offer lifecycle configuration.

AWS Storage Gateway - AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. All data transferred between the gateway and AWS storage is encrypted using SSL (for all three types of gateways - File, Volume and Tape Gateways). AWS Storage Gateway does not offer lifecycle configuration.

Reference:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/object-lifecycle-mgmt.html>

Domain

Technology

Question 47Incorrect

Which AWS service can be used to set up billing alarms to monitor estimated charges on your AWS account?

AWS Organizations

Correct answer

Amazon CloudWatch

AWS CloudTrail

Your answer is incorrect

AWS Cost Explorer

Overall explanation

Correct option:

Amazon CloudWatch

Amazon CloudWatch can be used to create an alarm to monitor your estimated charges. When you enable the monitoring of estimated charges for your AWS account, the estimated charges are calculated and sent several times daily to CloudWatch as metric data. You can choose to receive alerts by email when charges have exceeded a certain threshold. These alerts are triggered by Amazon CloudWatch and messages are sent using Amazon Simple Notification Service (Amazon SNS). Billing metric data is stored in the US East (N. Virginia) Region and reflects worldwide charges.

The alarm triggers when your account billing exceeds the threshold you specify. It triggers only when actual billing exceeds the threshold. It doesn't use projections based on your usage so far in the month.

Amazon CloudWatch Billing Alarms Overview:

Creating a Billing Alarm to Monitor Your Estimated AWS Charges

[PDF](#) | [Kindle](#) | [RSS](#)

You can monitor your estimated AWS charges by using Amazon CloudWatch. When you enable the monitoring of estimated charges for your AWS account, the estimated charges are calculated and sent several times daily to CloudWatch as metric data.

Billing metric data is stored in the US East (N. Virginia) Region and represents worldwide charges. This data includes the estimated charges for every service in AWS that you use, in addition to the estimated overall total of your AWS charges.

The alarm triggers when your account billing exceeds the threshold you specify. It triggers only when actual billing exceeds the threshold. It doesn't use projections based on your usage so far in the month.

If you create a billing alarm at a time when your charges have already exceeded the threshold, the alarm goes to the ALARM state immediately.

via -

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/monitor_estimated_charges_with_cloudwatch.html

Exam Alert:

It is useful to note the difference between Amazon CloudWatch Billing vs AWS Budgets:

Amazon CloudWatch Billing Alarms: Sends an alarm when the actual cost exceeds a certain threshold.

AWS Budgets: Sends an alarm when the actual cost exceeds the budgeted amount or even when the cost forecast exceeds the budgeted amount.

Incorrect options:

AWS CloudTrail - AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. With AWS CloudTrail, you can log, continuously monitor, and retain account activity related to actions across your AWS infrastructure. Billing alarms cannot be triggered via AWS CloudTrail.

AWS Organizations - AWS Organizations is an account management service that enables you to consolidate multiple AWS accounts into an organization that you create and centrally manage. Consolidated billing is a feature of AWS Organizations.

You can use the master account of your organization to consolidate and pay for all member accounts. Billing alarms cannot, however, be triggered using Consolidated Billing.

AWS Cost Explorer - AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer will help analyze your data at a high level or dive deeper into your cost and usage data using various reports (Monthly costs by AWS service, hourly and resource Level cost). Billing alarms cannot be triggered via AWS Cost Explorer.

Reference:

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/monitor_estimated_charges_with_cloudwatch.html

Domain

Billing and Pricing

Question 48Incorrect

Which of the following AWS authentication mechanisms supports an AWS Multi-Factor Authentication (AWS MFA) device that you can plug into a USB port on your computer?

Virtual Multi-Factor Authentication (AWS MFA) device

Hardware Multi-Factor Authentication (AWS MFA) device

Your answer is incorrect

SMS text message-based Multi-Factor Authentication (AWS MFA)

Correct answer

U2F security key

Overall explanation

Correct option:

U2F security key

Universal 2nd Factor (U2F) Security Key is a device that you can plug into a USB port on your computer. U2F is an open authentication standard hosted by the FIDO Alliance. When you enable a U2F security key, you sign in by entering your credentials and then tapping the device instead of manually entering a code.

How to enable the U2F Security Key for your own IAM user:

To enable a U2F security key for your own IAM user (console)

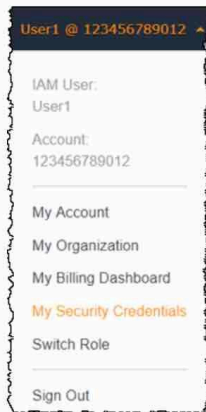
1. Use your AWS account ID or account alias, your IAM user name, and your password to sign in to the [IAM console](#).

Note

For your convenience, the AWS sign-in page uses a browser cookie to remember your IAM user name and account information. If you previously signed in as a different user, choose **Sign in to a different account** near the bottom of the page to return to the main sign-in page. From there, you can type your AWS account ID or account alias to be redirected to the IAM user sign-in page for your account.

To get your AWS account ID, contact your administrator.

2. In the navigation bar on the upper right, choose your user name, and then choose **My Security Credentials**.



3. On the **AWS IAM credentials** tab, in the **Multi-factor authentication** section, choose **Manage MFA device**.
4. In the **Manage MFA device** wizard, choose **U2F security key**, and then choose **Continue**.
5. Insert the U2F security key into your computer's USB port.



6. Tap the U2F security key, and then choose **Close** when U2F setup is complete.

via -

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa_enable_u2f.html

Incorrect options:

Virtual Multi-Factor Authentication (AWS MFA) device - This is a software app that runs on a phone or other device and emulates a physical device. The device generates a six-digit numeric code based upon a time-synchronized one-time password algorithm. The user must type a valid code from the device on a second webpage during sign-in. Each virtual MFA device assigned to a user must be unique.

Hardware Multi-Factor Authentication (AWS MFA) device - This is a hardware device that generates a six-digit numeric code based upon a time-synchronized one-time

password algorithm. The user must type a valid code from the device on a second webpage during sign-in. Each MFA device assigned to a user must be unique. A user cannot type a code from another user's device to be authenticated.

SMS text message-based Multi-Factor Authentication (AWS MFA) - This is a type of MFA in which the IAM user settings include the phone number of the user's SMS-compatible mobile device. When the user signs in, AWS sends a six-digit numeric code by SMS text message to the user's mobile device. The user is required to type that code on a second webpage during sign-in.

References:

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa_enable_u2f.html

Domain

Security and Compliance

Question 49

Incorrect
An e-commerce company wants to review the Payment Card Industry (PCI) reports on AWS Cloud. Which AWS resource can be used to address this use-case?

Correct answer

AWS Artifact

Your answer is incorrect

AWS Cost & Usage Report (AWS CUR)

AWS Trusted Advisor

AWS Secrets Manager

Overall explanation

Correct option:

AWS Artifact

AWS Artifact is your go-to, central resource for compliance-related information that matters to your organization. It provides on-demand access to AWS' security and compliance reports and select online agreements. Reports available in AWS Artifact include our Service Organization Control (SOC) reports, Payment Card Industry (PCI) reports, and certifications from accreditation bodies across geographies and compliance verticals that validate the implementation and operating effectiveness of

AWS security controls. It is not a service, it's a no-cost, self-service portal for on-demand access to AWS' compliance reports.

Incorrect options:

AWS Trusted Advisor - AWS Trusted Advisor is an online tool that provides you real-time guidance to help you provision your resources following AWS best practices. Whether establishing new workflows, developing applications, or as part of ongoing improvement, recommendations provided by Trusted Advisor regularly help keep your solutions provisioned optimally.

AWS Secrets Manager - AWS Secrets Manager helps you protect secrets needed to access your applications, services, and IT resources. The service enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. Users and applications retrieve secrets with a call to AWS Secrets Manager APIs, eliminating the need to hardcode sensitive information in plain text.

AWS Cost & Usage Report (AWS CUR) - The AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can use AWS Cost & Usage Report (AWS CUR) to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself. AWS updates the report in your bucket once a day in comma-separated value (CSV) format.

Reference:

<https://aws.amazon.com/artifact/>

Domain

Security and Compliance

Question 50Correct

A digital media company wants to convert English language subtitles into Spanish language subtitles. As a Cloud Practitioner, which AWS service would you recommend for this use-case?

Amazon Transcribe

Amazon Rekognition

Your answer is correct

Amazon Translate

Amazon Polly

Overall explanation

Correct option:

Amazon Translate

Amazon Translate is a neural machine translation service that delivers fast, high-quality, and affordable language translation. Amazon Translate allows you to localize content - such as websites and applications - for international users, and to easily translate large volumes of text efficiently.

Incorrect options:

Amazon Polly - You can use Amazon Polly to turn text into lifelike speech thereby allowing you to create applications that talk. Polly's Text-to-Speech (TTS) service uses advanced deep learning technologies to synthesize natural sounding human speech.

Amazon Transcribe - You can use Amazon Transcribe to add speech-to-text capability to your applications. Amazon Transcribe uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately. Amazon Transcribe can be used to transcribe customer service calls, to automate closed captioning and subtitling, and to generate metadata for media assets.

Amazon Rekognition - With Amazon Rekognition, you can identify objects, people, text, scenes, and activities in images and videos, as well as to detect any inappropriate content. Amazon Rekognition also provides highly accurate facial analysis and facial search capabilities that you can use to detect, analyze, and compare faces for a wide variety of user verification, people counting, and public safety use cases.

Reference:

<https://aws.amazon.com/translate/>

Domain

Technology

Question 51Correct

Which entity ensures that your application on Amazon Elastic Compute Cloud (Amazon EC2) always has the right amount of capacity to handle the current traffic demand?

Your answer is correct

Amazon EC2 Auto Scaling

Multi-AZ deployment

Network Load Balancer

Application Load Balancer

Overall explanation

Correct option:

Amazon EC2 Auto Scaling

Amazon EC2 Auto Scaling helps you ensure that you have the correct number of Amazon EC2 instances available to handle the load for your application. You create collections of Amazon EC2 instances, called Auto Scaling groups. You can specify the minimum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes below this size.

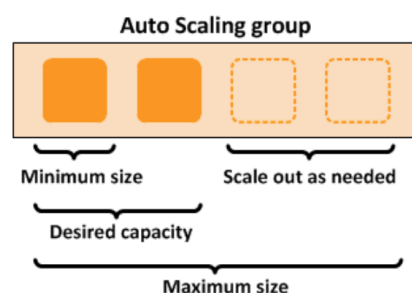
Amazon EC2 Auto Scaling Overview:

What Is Amazon EC2 Auto Scaling?

[PDF](#) | [Kindle](#) | [RSS](#)

Amazon EC2 Auto Scaling helps you ensure that you have the correct number of Amazon EC2 instances available to handle the load for your application. You create collections of EC2 instances, called *Auto Scaling groups*. You can specify the minimum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes below this size. You can specify the maximum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes above this size. If you specify the desired capacity, either when you create the group or at any time thereafter, Amazon EC2 Auto Scaling ensures that your group has this many instances. If you specify scaling policies, then Amazon EC2 Auto Scaling can launch or terminate instances as demand on your application increases or decreases.

For example, the following Auto Scaling group has a minimum size of one instance, a desired capacity of two instances, and a maximum size of four instances. The scaling policies that you define adjust the number of instances, within your minimum and maximum number of instances, based on the criteria that you specify.



via -

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/what-is-amazon-ec2-auto-scaling.html>

Incorrect options:

Multi-AZ deployment - With Availability Zones (AZ), you can design and operate applications and databases that automatically failover between zones without interruption. Multi-AZ deployment of Amazon EC2 instances provided high availability, it does not help in scaling resources.

Network Load Balancer - Network Load Balancer is best suited for load balancing of Transmission Control Protocol (TCP), User Datagram Protocol (UDP) and Transport Layer Security (TLS) traffic where extreme performance is required. It distributes traffic, does not scale resources.

Application Load Balancer - An Application Load Balancer serves as the single point of contact for clients. The load balancer distributes incoming application traffic across multiple targets, such as EC2 instances, in multiple Availability Zones. It distributes traffic, does not scale resources.

Reference:

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/what-is-amazon-ec2-auto-scaling.html>

Domain

Cloud Concepts

Question 52

Which of the following types are free under the Amazon Simple Storage Service (Amazon S3) pricing model? (Select two)

Data storage fee for objects stored in S3 Glacier

Your selection is correct

Data transferred out to an Amazon Elastic Compute Cloud (Amazon EC2) instance, when the instance is in the same AWS Region as the S3 bucket

Data transferred out to an Amazon Elastic Compute Cloud (Amazon EC2) instance in any AWS Region

Correct selection

Data transferred in from the internet

Your selection is incorrect

Data storage fee for objects stored in S3 Standard

Overall explanation

Correct options:

Data transferred in from the internet

Data transferred out to an Amazon Elastic Compute Cloud (Amazon EC2) instance, when the instance is in the same AWS Region as the S3 bucket

There are four cost components to consider for S3 pricing – storage pricing; request and data retrieval pricing; data transfer and transfer acceleration pricing; and data management features pricing. Under "Data Transfer", You pay for all bandwidth into and out of Amazon S3, except for the following: (1) Data transferred in from the internet, (2) Data transferred out to an Amazon Elastic Compute Cloud (Amazon EC2) instance, when the instance is in the same AWS Region as the S3 bucket, (3) Data transferred out to Amazon CloudFront (CloudFront).

Incorrect options:

Data transferred out to an Amazon Elastic Compute Cloud (Amazon EC2) instance in any AWS Region - This is incorrect. Data transfer charges apply when the instance is not in the same AWS Region as the S3 bucket.

Data storage fee for objects stored in S3 Standard - S3 Standard charges a storage fee for objects.

Data storage fee for objects stored in S3 Glacier - S3 Glacier charges a storage fee for objects.

Reference:

<https://aws.amazon.com/s3/pricing/>

Domain

Billing and Pricing

Question 53Correct

Which of the following entities are part of an Amazon Virtual Private Cloud (Amazon VPC) in the AWS Cloud? (Select two)

Your selection is correct

Internet Gateway

AWS Storage Gateway

Your selection is correct

Subnet

API Gateway

Object

Overall explanation

Correct options:

Subnet

Internet Gateway

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that you've defined.

The following are the key concepts for VPCs:

Virtual private cloud (VPC) — A virtual network dedicated to your AWS account.

Subnet — A range of IP addresses in your VPC.

Route table — A set of rules, called routes, that are used to determine where network traffic is directed.

Internet Gateway — A gateway that you attach to your VPC to enable communication between resources in your VPC and the internet.

VPC endpoint — Enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection.

Incorrect options:

AWS Storage Gateway - AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage. Customers use Storage Gateway to simplify storage management and reduce costs for key hybrid cloud storage use cases. Storage Gateway is not part of VPC.

API Gateway - Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale. APIs act as the "front door" for applications to access data, business logic, or functionality from your backend services. API Gateway is not part of a VPC.

Object - Buckets and objects are part of Amazon S3. Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Reference:

https://docs.amazonaws.cn/en_us/vpc/latest/userguide/what-is-amazon-vpc.html

Domain

Cloud Concepts

Question 54Incorrect

Amazon Elastic Compute Cloud (Amazon EC2) Spot instances are a best-fit for which of the following scenarios?

Correct answer

To run any containerized workload with Amazon Elastic Container Service (Amazon ECS) that can be interrupted

To install cost-effective Amazon RDS database

Your answer is incorrect

To run batch processes for critical workloads

To run scheduled jobs (jobs that run at the same time every day)

Overall explanation

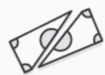
Correct option:

To run any containerized workload with Amazon Elastic Container Service (Amazon ECS) that can be interrupted

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud. Spot Instances are available at up to a 90% discount compared to On-Demand prices.

Containers are stateless, fault-tolerant and a great fit for Spot Instances. Spot Instances can be used with Elastic Container Service (ECS) or Elastic Container Service for Kubernetes (EKS) to run any containerized workload, from distributed parallel test systems to applications that map millions of miles a day. Spot instances provide the flexibility of ad-hoc provisioning for multiple instance types in different Availability Zones, with an option to hibernate, stop or terminate instances when EC2 needs the capacity back and Spot Instances are reclaimed.

Benefits of Using Spot Instances to Run Containers



LOWER COSTS

Significant price savings of up to 90% over On-Demand EC2 Instances. Simply mix Spot Instances with On-Demand and/or Reserved Instances or run on 100% Spot Instances to optimize cost and performance.



FASTER RESULTS

Easily run multiple projects simultaneously and speed up job flows to generate business results faster and innovate faster without breaking the bank. Run and scale to large numbers of parallel tasks via Spot Fleet or Spot Instance pool.



RESOURCE FLEXIBILITY

Flexibility of ad-hoc provisioning for multiple instance types in different Availability Zones, with an option to hibernate, stop or terminate instances when EC2 needs the capacity back and Spot Instances are reclaimed.



EASE OF USE

Easily launch a Spot Instance via an API call, EC2 Fleet and the AWS Management Console. EC2 Spot is also integrated with other AWS services such as ECS, EKS, EC2 Auto Scaling groups and CloudFormation.

via - <https://aws.amazon.com/ec2/spot/containers-for-less/>

Incorrect options:

To install cost-effective Amazon RDS database - Spot instance capacity allocated to you can be taken back anytime without notice if AWS needs them. Hence, Spot instances can only be used as additional compute capacity and not for hosting or installing any software or database.

To run batch processes for critical workloads - Business-critical workloads cannot be run on Spot instances.

To run scheduled jobs (jobs that run at the same time every day) - There is no guarantee that a Spot instance will be available at a specific time every day. For a scheduled requirement, Scheduled Reserved instances should be used.

Reference:

<https://aws.amazon.com/ec2/spot/containers-for-less/>

Domain

Technology

Question 55Incorrect

Which AWS entity enables you to privately connect your Amazon Virtual Private Cloud (Amazon VPC) to an Amazon Simple Queue Service (Amazon SQS) queue?

Internet Gateway

Your answer is incorrect

VPC Gateway Endpoint

AWS Direct Connect

Correct answer

VPC Interface Endpoint

Overall explanation

Correct option:

VPC Interface Endpoint

An interface endpoint is an elastic network interface (ENI) with a private IP address from the IP address range of your subnet that serves as an entry point for traffic destined to a supported service. Interface endpoints are powered by AWS PrivateLink, a technology that enables you to privately access services by using private IP addresses. AWS PrivateLink restricts all network traffic between your VPC and services to the Amazon network. You do not need an internet gateway, a Network Address Translation (NAT) device, or a virtual private gateway.

Exam Alert:

You may see a question around this concept in the exam. Just remember that only Amazon S3 and Amazon DynamoDB support VPC gateway endpoint. All other services that support VPC Endpoints use a VPC interface endpoint (note that Amazon S3 supports the VPC interface endpoint as well).

Incorrect options:

VPC Gateway Endpoint - A Gateway Endpoint is a gateway that you specify as a target for a route in your route table for traffic destined to a supported AWS service. The following AWS services are supported: Amazon S3, DynamoDB. You cannot use VPC Gateway Endpoint to privately connect your VPC to an Amazon SQS queue.

AWS Direct Connect - AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. You can use AWS Direct Connect to establish a private virtual interface from your on-premise network directly to your Amazon VPC. This private connection takes at least one month for completion. You cannot use AWS Direct Connect to privately connect your VPC to an Amazon SQS queue.

Internet Gateway - An Internet Gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between your VPC and the internet. An internet gateway serves two purposes: to provide a target in your VPC route tables for internet-routable traffic, and to perform network address translation (NAT) for instances that have been assigned public IPv4 addresses. You cannot use an Internet Gateway to privately connect your VPC to an Amazon SQS queue.

Reference:

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-endpoints.html>

Domain

Security and Compliance

Question 56Correct

Which AWS service will you use to provision the same AWS infrastructure across multiple AWS accounts and regions?

AWS OpsWorks

AWS CodeDeploy

AWS Systems Manager

Your answer is correct

AWS CloudFormation

Overall explanation

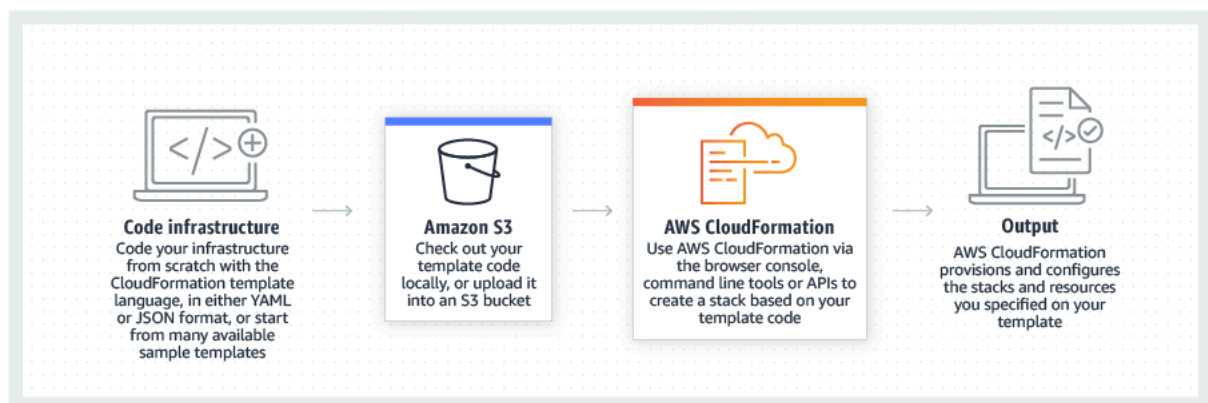
Correct option:

AWS CloudFormation

AWS CloudFormation allows you to use programming languages or a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all Regions and accounts. A stack is a collection of AWS resources that you can manage as a single unit. In other words, you can create, update, or delete a collection of resources by creating, updating, or deleting stacks.

AWS CloudFormation StackSets extends the functionality of stacks by enabling you to create, update, or delete stacks across multiple accounts and regions with a single operation. Using an administrator account, you define and manage an AWS CloudFormation template, and use the template as the basis for provisioning stacks into selected target accounts across specified regions.

How CloudFormation Works:



via - <https://aws.amazon.com/cloudformation/>

Incorrect options:

AWS CodeDeploy - AWS CodeDeploy is a fully managed deployment service that automates software deployments to a variety of compute services such as Amazon EC2, AWS Fargate, AWS Lambda, and your on-premises servers. AWS CodeDeploy makes it easier for you to rapidly release new features, helps you avoid downtime during application deployment, and handles the complexity of updating your applications. You cannot use this service to provision AWS infrastructure.

AWS OpsWorks - AWS OpsWorks is a configuration management service that provides managed instances of Chef and Puppet. OpsWorks lets you use Chef and Puppet to automate how servers are configured, deployed and managed across your Amazon EC2 instances or on-premises compute environments. You cannot use

OpsWorks for running commands or managing patches on servers. You cannot use this service to provision AWS infrastructure.

AWS Systems Manager - AWS Systems Manager gives you visibility and control of your infrastructure on AWS. Systems Manager provides a unified user interface so you can view operational data from multiple AWS services and allows you to automate operational tasks across your AWS resources. With Systems Manager, you can group resources, like Amazon EC2 instances, Amazon S3 buckets, or Amazon RDS instances, by application, view operational data for monitoring and troubleshooting, and take action on your groups of resources. You cannot use this service to provision AWS infrastructure.

Reference:

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/what-is-cfnstacks.html>

Domain

Technology

Question 57Incorrect

Which AWS service would you use to create a logically isolated section of the AWS Cloud where you can launch AWS resources in your virtual network?

Correct answer

Virtual private Cloud (VPC)

Your answer is incorrect

Virtual Private Network (VPN)

Subnet

Network Access Control List (Network ACL)

Overall explanation

Correct option:

Virtual private Cloud (VPC)

Amazon Virtual Private Cloud (Amazon VPC) is a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including the selection of your IP address range, creation of subnets, and configuration of route tables and network gateways. You can easily customize the network configuration of your Amazon VPC using public and private subnets.

Incorrect options:

Virtual Private Network (VPN) - AWS Virtual Private Network (AWS VPN) lets you establish a secure and private encrypted tunnel from your on-premises network to the AWS global network. AWS VPN is comprised of two services: AWS Site-to-Site VPN and AWS Client VPN. You cannot use VPN to create a logically isolated section of the AWS Cloud.

Subnet - A subnet is a range of IP addresses within your VPC. A subnet is not an AWS service, so this option is ruled out.

Network Access Control List (Network ACL) - A network access control list (network NACL) is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets. A network access control list (network NACL) is not an AWS service, so this option is ruled out.

Reference:

<https://aws.amazon.com/vpc/>

Domain

Technology

Question 58Correct

A media company uploads its media (audio and video) files to a centralized Amazon Simple Storage Service (Amazon S3) bucket from geographically dispersed locations. Which of the following solutions can the company use to optimize transfer speeds?

Amazon CloudFront

AWS Direct Connect

Your answer is correct

Amazon S3 Transfer Acceleration (S3TA)

AWS Global Accelerator

Overall explanation

Correct option:

Amazon S3 Transfer Acceleration (S3TA)

Amazon S3 Transfer Acceleration (S3TA) enables fast, easy, and secure transfers of files over long distances between your client and your Amazon S3 bucket. S3 Transfer Acceleration leverages Amazon CloudFront's globally distributed AWS Edge

Locations. As data arrives at an AWS Edge Location, data is routed to your Amazon S3 bucket over an optimized network path. S3 Transfer Acceleration is designed to optimize transfer speeds from across the world into S3 buckets. If you are uploading to a centralized bucket from geographically dispersed locations, or if you regularly transfer GBs or TBs of data across continents, you may save hours or days of data transfer time with S3 Transfer Acceleration.

Amazon S3 Transfer Acceleration (S3TA) benefits:

Move data faster over long distances	Reduce network variability	Shorten the distance to S3	Maximize bandwidth utilization
S3TA can accelerate long-distance transfers to and from your Amazon S3 buckets. The longer the distance between your client application (mobile, web application, or upload tool) and the target S3 bucket, the more S3TA can help. And if S3TA would not accelerate a transfer, you are not charged.	For applications interacting with your S3 buckets through the S3 API from outside of your bucket's region, S3TA helps avoid the variability in Internet routing and congestion. It does this by routing your uploads and downloads over the AWS global network infrastructure, so you get the benefit of our network optimizations.	S3TA shortens the distance between client applications and AWS servers that acknowledge PUTS and GETS to Amazon S3 using our global network of hundreds of CloudFront Edge Locations. We automatically route your uploads and downloads through the closest Edge Locations to your application.	S3TA on average fully utilizes your bandwidth for transfers, and minimizes the effect of distance on throughput. This helps to ensure consistently fast performance to Amazon S3 regardless of your client's location.

via - <https://aws.amazon.com/s3/transfer-acceleration/>

Incorrect options:

Amazon CloudFront - Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency, high transfer speeds, all within a developer-friendly environment. CloudFront is used for content delivery than for data uploads. CloudFront caches data and a subsequent request for a webpage will not go to the origin server, but will be served from the cache. S3 Transfer Acceleration is a better option for the given use-case.

AWS Direct Connect - AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. You can use AWS Direct Connect to establish a private virtual interface from your on-premise network directly to your Amazon VPC. This private connection takes at least one month for completion. You cannot use Direct Connect to optimize media uploads into S3.

AWS Global Accelerator - AWS Global Accelerator is a service that improves the availability and performance of your applications with local or global users. It provides static IP addresses that act as a fixed entry point to your application endpoints in a single or multiple AWS Regions, such as your Application Load

Balancers, Network Load Balancers or Amazon EC2 instances. Similar to CloudFront it uses AWS Global network and edge locations for enhanced performance. It's an overall performance enhancer than an upload speed accelerator. You cannot use Global Accelerator to optimize media uploads into S3.

Reference:

<https://aws.amazon.com/s3/transfer-acceleration/>

Domain

Technology

Question 59Correct

Which Amazon Route 53 routing policy would you use to route traffic to a single resource such as a web server for your website?

Failover routing

Weighted routing

Your answer is correct

Simple routing

Latency-based routing

Overall explanation

Correct option:

Simple routing

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service. It is designed to give developers and businesses an extremely reliable and cost-effective way to route end users to Internet applications by translating names like `www.example.com` into the numeric IP addresses like `192.0.2.1` that computers use to connect to each other.

Simple routing lets you configure standard DNS records, with no special Amazon Route 53 routing such as weighted or latency. With simple routing, you typically route traffic to a single resource, for example, to a web server for your website.

Amazon Route 53 Routing Policy Overview:

Choosing a routing policy

[PDF](#) | [Kindle](#) | [RSS](#)

When you create a record, you choose a routing policy, which determines how Amazon Route 53 responds to queries:

- **Simple routing policy** – Use for a single resource that performs a given function for your domain, for example, a web server that serves content for the example.com website.
- **Failover routing policy** – Use when you want to configure active-passive failover.
- **Geolocation routing policy** – Use when you want to route traffic based on the location of your users.
- **Geoproximity routing policy** – Use when you want to route traffic based on the location of your resources and, optionally, shift traffic from resources in one location to resources in another.
- **Latency routing policy** – Use when you have resources in multiple AWS Regions and you want to route traffic to the region that provides the best latency.
- **Multivalue answer routing policy** – Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random.
- **Weighted routing policy** – Use to route traffic to multiple resources in proportions that you specify.

via -

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>

Incorrect options:

Failover routing - This routing policy is used when you want to configure active-passive failover.

Weighted routing - This routing policy is used to route traffic to multiple resources in proportions that you specify.

Latency-based routing - This routing policy is used when you have resources in multiple AWS Regions and you want to route traffic to the region that provides the best latency.

Reference:

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html>

Domain

Technology

Question 60Correct

Which of the following can you use to run a bootstrap script while launching an Amazon Elastic Compute Cloud (Amazon EC2) instance?

Amazon EC2 instance AMI data

Your answer is correct

Amazon EC2 instance user data

Amazon EC2 instance configuration data

Amazon EC2 instance metadata

Overall explanation

Correct option:

Amazon EC2 instance user data

Amazon EC2 instance user data is the data that you specified in the form of a bootstrap script or configuration parameters while launching your instance.

Amazon EC2 instance metadata and user data:

Instance metadata and user data

PDF | Kindle | RSS

Instance metadata is data about your instance that you can use to configure or manage the running instance. Instance metadata is divided into categories, for example, host name, events, and security groups.

You can also use instance metadata to access *user data* that you specified when launching your instance. For example, you can specify parameters for configuring your instance, or include a simple script. You can build generic AMIs and use user data to modify the configuration files supplied at launch time.

For example, if you run web servers for various small businesses, they can all use the same generic AMI and retrieve their content from the Amazon S3 bucket that you specify in the user data at launch. To add a new customer at any time, create a bucket for the customer, add their content, and launch your AMI with the unique bucket name provided to your code in the user data. If you launch more than one instance at the same time, the user data is available to all instances in that reservation. Each instance that is part of the same reservation has a unique `ami-launch-index` number, allowing you to write code that controls what to do. For example, the first host might elect itself as an initial master node in a cluster. For a detailed AMI launch example, see [Example: AMI launch index value](#).

via -

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-instance-metadata.html>

Incorrect options:

Amazon EC2 instance metadata - Amazon EC2 instance metadata is data about your instance that you can use to manage the instance. You can get instance items such as `ami-id`, `public-hostname`, `local-hostname`, `hostname`, `public-ipv4`, `local-ipv4`, `public-keys`, `instance-id` by using instance metadata. You cannot use Amazon EC2 instance metadata to run a bootstrap script while launching an Amazon EC2 instance. So this option is incorrect.

Amazon EC2 instance configuration data

Amazon EC2 instance AMI data

There is no such thing as Amazon EC2 instance configuration data or Amazon EC2 instance AMI data. These options have been added as distractors.

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-instance-metadata.html>

Domain

Technology

Question 61 Correct

The DevOps team at a Big Data consultancy has set up Amazon Elastic Compute Cloud (Amazon EC2) instances across two AWS Regions for its flagship application. Which of the following characterizes this application architecture?

Deploying the application across two AWS Regions improves security

Deploying the application across two AWS Regions improves agility

Deploying the application across two AWS Regions improves scalability

Your answer is correct

Deploying the application across two AWS Regions improves availability

Overall explanation

Correct option:

Deploying the application across two AWS Regions improves availability

Highly available systems are those that can withstand some measure of degradation while remaining available. Each AWS Region is fully isolated and comprised of multiple Availability Zones (AZ), which are fully isolated partitions of AWS infrastructure. To better isolate any issues and achieve high availability, you can partition applications across multiple Availability Zones (AZ) in the same AWS Region or even across multiple AWS Regions.

Key Benefits of AWS Global Infrastructure:

Security Security at AWS starts with our core infrastructure. Custom-built for the cloud and designed to meet the most stringent security requirements in the world, our infrastructure is monitored 24/7 to help ensure the confidentiality, integrity, and availability of your data. All data flowing across the AWS global network that interconnects our datacenters and Regions is automatically encrypted at the physical layer before it leaves our secured facilities. You can build on the most secure global infrastructure, knowing you always control your data, including the ability to encrypt it, move it, and manage retention at any time.	Availability AWS delivers the highest network availability of any cloud provider, with 7x fewer down time hours than the next largest cloud provider.* Each region is fully isolated and comprised of multiple AZ's, which are fully isolated partitions of our infrastructure. To better isolate any issues and achieve high availability, you can partition applications across multiple AZ's in the same region. In addition, AWS control planes and the AWS management console are distributed across regions, and include regional API endpoints, which are designed to operate securely for at least 24 hours if isolated from the global control plane functions without requiring customers to access the region or its API endpoints via external networks during any isolation.	Performance The AWS Global Infrastructure is built for performance. AWS Regions offer low latency, low packet loss, and high overall network quality. This is achieved with a fully redundant 100 GbE fiber network backbone, often providing many terabits of capacity between Regions. AWS Local Zones and AWS Wavelength, with our telco providers, provide performance for applications that require single-digit millisecond latencies by delivering AWS infrastructure and services closer to end-users and 5G connected devices. Whatever your application needs, you can quickly spin up resources as you need them, deploying hundreds or even thousands of servers in minutes.
Global Footprint AWS has the largest global infrastructure footprint of any provider, and this footprint is constantly increasing at a significant rate. When deploying your applications and workloads to the cloud, you have the flexibility in selecting a technology infrastructure that is closest to your primary target of users. You can run your workloads on the cloud that delivers the best support for the broadest set of applications, even those with the highest throughput and lowest latency requirements. And if your data lives off this planet, you can use AWS Ground Station, which provides satellite antennas in close proximity to AWS infrastructure Regions.	Scalability The AWS Global Infrastructure enables companies to be extremely flexible and take advantage of the conceptually infinite scalability of the cloud. Customers used to over provision to ensure they had enough capacity to handle their business operations at the peak level of activity. Now, they can provision the amount of resources that they actually need, knowing they can instantly scale up or down along with the needs of their business, which also reduces cost and improves the customer's ability to meet their user's demands. Companies can quickly spin up resources as they need them, deploying hundreds or even thousands of servers in minutes.	Flexibility The AWS Global Infrastructure gives you the flexibility of choosing how and where you want to run your workloads, and when you do you are using the same network, control plane, API's, and AWS services. If you would like to run your applications globally you can choose from any of the AWS Regions and AZ's. If you need to run your applications with single-digit millisecond latencies to mobile devices and end-users you can choose AWS Local Zones or AWS Wavelength. Or if you would like to run your applications on-premises you can choose AWS Outposts.

via - <https://aws.amazon.com/about-aws/global-infrastructure/>

Incorrect options:

Deploying the application across two AWS Regions improves agility - Agility refers to the ability of the cloud to give you easy access to a broad range of technologies so that you can innovate faster and build nearly anything that you can imagine. You can quickly spin up resources as you need them – from infrastructure services, such as compute, storage, and databases, to Internet of Things, machine learning, data lakes and analytics, and much more. Deploying the application across two AWS Regions does not improve agility.

Deploying the application across two AWS Regions improves security - The application security is dependent on multiple factors such as data encryption, IAM policies, IAM roles, virtual private cloud (VPC) security configurations, security groups, network access control lists (network ACL), etc. Deploying the application across two AWS Regions directly impacts availability. So this option is not the best fit for the given use-case.

Deploying the application across two AWS Regions improves scalability - For the given use-case, you can improve the scalability of the application by using an Application Load Balancer with an Auto Scaling group. Deploying the application

across two AWS Regions directly impacts availability. So this option is not the best fit for the given use-case.

Reference:

<https://aws.amazon.com/about-aws/global-infrastructure/>

Domain

Cloud Concepts

Question 62Correct

A financial services company wants to migrate from its on-premises data center to AWS Cloud. As a Cloud Practitioner, which AWS service would you recommend so that the company can compare the cost of running their IT infrastructure on-premises vs AWS Cloud?

AWS Cost Explorer

AWS Trusted Advisor

Your answer is correct

AWS Pricing Calculator

AWS Budgets

Overall explanation

Correct option:

AWS Pricing Calculator

AWS Pricing Calculator lets you explore AWS services and create an estimate for the cost of your use cases on AWS. You can model your solutions before building them, explore the price points and calculations behind your estimate, and find the available instance types and contract terms that meet your needs. This enables you to make informed decisions about using AWS. You can plan your AWS costs and usage or price out setting up a new set of instances and services. AWS Pricing Calculator can be accessed at <https://calculator.aws/#/>.

AWS also offers a complimentary service called Migration Evaluator (Formerly TSO Logic) to create data-driven business cases for AWS Cloud planning and migration.

Incorrect options:

AWS Trusted Advisor - AWS Trusted Advisor provides recommendations that help you follow AWS best practices. Trusted Advisor evaluates your account by using checks. These checks identify ways to optimize your AWS infrastructure, improve

security and performance, reduce costs, and monitor service quotas. This service cannot be used to compare the cost of running the IT infrastructure on-premises vs AWS Cloud.

AWS Cost Explorer - AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown of all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends. AWS Cost Explorer cannot be used to compare the cost of running the IT infrastructure on-premises vs AWS Cloud.

AWS Budgets - AWS Budgets gives the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. Budgets can be created at the monthly, quarterly, or yearly level, and you can customize the start and end dates. You can further refine your budget to track costs associated with multiple dimensions, such as AWS service, linked account, tag, and others. AWS Budgets cannot be used to compare the cost of running the IT infrastructure on-premises vs AWS Cloud.

Reference:

<https://calculator.aws/#/>

<https://docs.aws.amazon.com/whitepapers/latest/how-aws-pricing-works/aws-pricing-tools.html>

<https://aws.amazon.com/migration-evaluator/>

Domain

Billing and Pricing

Question 63Correct

AWS Marketplace facilitates which of the following use-cases? (Select two)

Your selection is correct

AWS customer can buy software that has been bundled into customized Amazon Machine Image (AMIs) by the AWS Marketplace sellers

Purchase compliance documents from third-party vendors

Buy Amazon EC2 Standard Reserved Instances (RI)

Your selection is correct

Sell Software as a Service (SaaS) solutions to AWS customers

Raise request for purchasing AWS Direct Connect connection

Overall explanation

Correct option:

Sell Software as a Service (SaaS) solutions to AWS customers

AWS customer can buy software that has been bundled into customized Amazon Machine Image (AMIs) by the AWS Marketplace sellers

AWS Marketplace is a digital catalog with thousands of software listings from independent software vendors that make it easy to find, test, buy, and deploy software that runs on AWS. The AWS Marketplace enables qualified partners to market and sell their software to AWS Customers.

AWS Marketplace offers two ways for sellers to deliver software to customers: Amazon Machine Image (AMI) and Software as a Service (SaaS).

Amazon Machine Image (AMI): Offering an AMI is the preferred option for listing products in AWS Marketplace. Partners have the option for free or paid products. Partners can offer paid products charged by the hour or month.

Bring-Your-Own-License (BYOL) is also available and enables customers with existing software licenses to easily migrate to AWS.

Software as a Service (SaaS): If you offer a SaaS solution running on AWS (and are unable to build your product into an AMI) the SaaS listing offers our partners a way to market their software to customers.

Incorrect options:

Purchase compliance documents from third-party vendors - There is no third party vendor for providing compliance documents. AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to AWS' security and compliance reports and select online agreements.

Buy Amazon EC2 Standard Reserved Instances (RI) - Amazon EC2 Standard

Reserved Instances (RI) can be bought from the Amazon EC2 console at

<https://console.aws.amazon.com/ec2/>

Raise request for purchasing AWS Direct Connect connection - AWS Direct Connect

connection can be raised from the AWS management console at

<https://console.aws.amazon.com/directconnect/v2/home>

References:

<https://aws.amazon.com/partners/aws-marketplace/>

<https://aws.amazon.com/artifact/>

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ri-market-concepts-buying.html#ri-queued-purchase>

Domain

Billing and Pricing

Question 64Incorrect

Which of the following statements are true regarding Amazon Simple Storage Service (Amazon S3) (Select two)?

Your selection is incorrect

Amazon Simple Storage Service (Amazon S3) is a fully managed, elastic file system storage service used as database backup

Correct selection

Amazon Simple Storage Service (Amazon S3) stores data in a flat non-hierarchical structure

Amazon Simple Storage Service (Amazon S3) is a block storage service designed for a broad range of workloads

Your selection is correct

Amazon Simple Storage Service (Amazon S3) is a key value based object storage service

You can install databases on Amazon Simple Storage Service (Amazon S3)

Overall explanation

Correct options:

Amazon Simple Storage Service (Amazon S3) is a key value based object storage service

Amazon Simple Storage Service (Amazon S3) stores data in a flat non-hierarchical structure

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Amazon Simple Storage Service (Amazon S3) stores data in a flat non-hierarchical structure. All objects are stored in Amazon S3 buckets and can be organized with shared names called prefixes. You can also append up to 10 key-value pairs called Amazon S3 object tags to each object, which can be created, updated, and deleted throughout an object's lifecycle.

Incorrect options:

Amazon Simple Storage Service (Amazon S3) is a block storage service designed for a broad range of workloads - Block storage service is provided by Amazon Elastic Block Store (Amazon EBS) to provide persistent block-level storage volumes for use with Amazon EC2 instances. Amazon S3 is an object storage service.

Amazon Simple Storage Service (Amazon S3) is a fully managed, elastic file system storage service used as database backup - Amazon Elastic File System (Amazon EFS) provides a simple, scalable, fully managed elastic NFS file system for use with AWS Cloud services and on-premises resources. Amazon S3 is an object storage service.

You can install databases on Amazon Simple Storage Service (Amazon S3) - Amazon S3 is an object storage service. You cannot install databases on Amazon S3.

Reference:

<https://aws.amazon.com/s3/features/>

Domain

Technology

Question 65Incorrect

Which of the following is a container service of AWS?

Your answer is incorrect

Amazon SageMaker

Amazon Simple Notification Service (Amazon SNS)

AWS Elastic Beanstalk

Correct answer

AWS Fargate

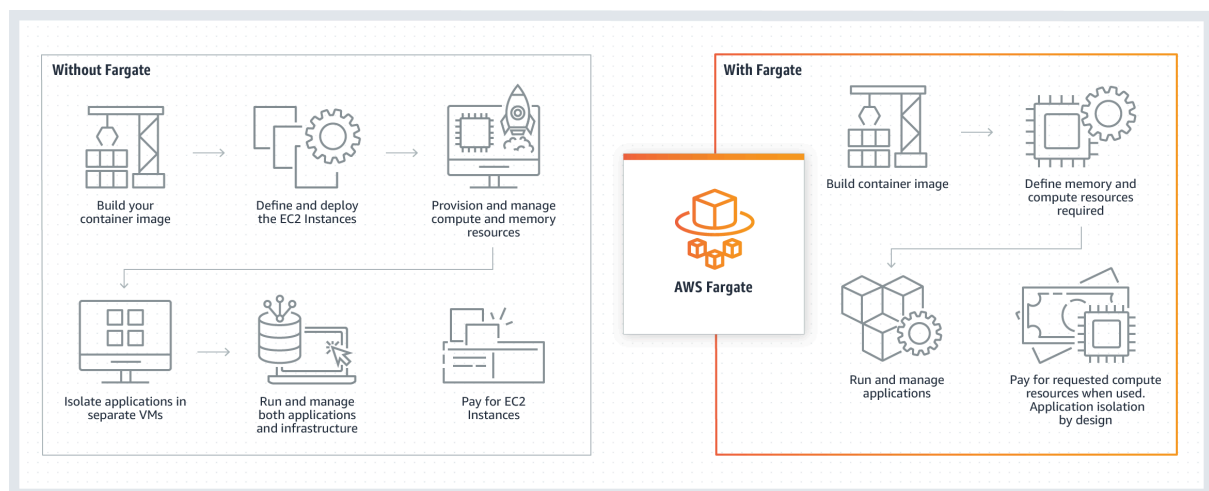
Overall explanation

Correct option:

AWS Fargate

AWS Fargate is a serverless compute engine for containers that works with both Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS). Fargate makes it easy for you to focus on building your applications. Fargate removes the need to provision and manage servers, lets you specify and pay for resources per application, and improves security through application isolation by design.

How AWS Fargate Works:



via - <https://aws.amazon.com/fargate/>

Incorrect options:

AWS Elastic Beanstalk - AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services. You simply upload your code and Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, auto-scaling to application health monitoring. Beanstalk provisions servers so it is not a serverless service.

Amazon Simple Notification Service (Amazon SNS) - Amazon Simple Notification Service (Amazon SNS) is a highly available, durable, secure, fully managed pub/sub

messaging service that enables you to decouple microservices, distributed systems, and serverless applications.

Amazon SageMaker - Amazon SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly. SageMaker removes the heavy lifting from each step of the machine learning process to make it easier to develop high-quality models.

Reference:

<https://aws.amazon.com/fargate/>